



Exploring User Acceptance of Scenic Area AR Real-Scene Map: The Role of System Quality, Information Quality, Immersion, and Perceived Enjoyment

Shuo Li, Yuhui Wang & Lei Liu

To cite this article: Shuo Li, Yuhui Wang & Lei Liu (09 Mar 2026): Exploring User Acceptance of Scenic Area AR Real-Scene Map: The Role of System Quality, Information Quality, Immersion, and Perceived Enjoyment, International Journal of Human-Computer Interaction, DOI: [10.1080/10447318.2026.2629518](https://doi.org/10.1080/10447318.2026.2629518)

To link to this article: <https://doi.org/10.1080/10447318.2026.2629518>



Published online: 09 Mar 2026.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)



Exploring User Acceptance of Scenic Area AR Real-Scene Map: The Role of System Quality, Information Quality, Immersion, and Perceived Enjoyment

Shuo Li , Yuhui Wang  and Lei Liu 

School of Architecture and Art, Central South University, Changsha, China

ABSTRACT

The tourism industry is undergoing rapid digital transformation, with augmented reality (AR) technologies increasingly integrated into tourism services. AR real-scene maps enhance the tourism experience, yet existing research focuses more on technology and less on user adoption. Based on TAM, this study introduces system quality, information quality, immersion, and perceived enjoyment to examine tourists' usage intention. Using questionnaire data from 322 respondents and SEM, findings show significant effects among key variables. Gender, age, and usage experience also play moderating roles. This study offers theoretical and practical implications for AR map design and tourism experience optimization.

KEYWORDS

Augmented reality (AR); AR real-scene map; technology acceptance model (TAM); user acceptance; tourism

1. Introduction

In recent years, the tourism industry has played an increasingly important role in global economic and social development, not only expanding in scale and market share but also contributing significantly to local economic growth and social progress (Briz, 2021; Postma et al., 2017). Driven by continuous technological advances and the digital economy, the digitalization of tourism has emerged as a global trend (Happ & Ivancsó-Horváth, 2018; Mussina et al., 2020). It has accelerated the integration of technologies such as artificial intelligence, virtual reality (VR), and augmented reality (AR) into tourism services to enhance the visitor experience and engagement (Cynthia et al., 2021).

The rapid advancement of mobile internet, smart devices, and augmented reality technologies has accelerated the digital transformation of geographic information services (Veenendaal et al., 2017), encouraging tourist destinations to improve visitor satisfaction and cultural dissemination through digital solutions (Abdel Rady & Khalf, 2019; Cardoso & Ruiz, 2021).

Within this broader process of digital tourism, various digital products—including reservation systems, intelligent voice guides, and information platforms—have become essential tools for improving service efficiency and user experience (Gajdošík, 2018). Among them, map systems based on geographic information technologies are particularly critical (Eboy, 2017). As an important component of the destination service system, maps not only help tourists obtain directional information and understand spatial structures in unfamiliar and complex attraction environments, but also guide wayfinding behavior, directly influencing the accuracy of route decisions and overall travel efficiency (Metz, 2022; Mieli et al., 2024). When map information is unclear or inaccurate, tourists are more likely to lose their way, which in turn undermines the overall tourism experience. Consequently, developing more intuitive and efficient map-based navigation systems to optimize route planning and information acquisition has become a core objective of digital services in tourist destinations, as such systems can enhance visitors' overall evaluations of tourism experience quality (Cabeleira & Vaz de Carvalho, 2025; Rizaludin et al., 2025).

However, traditional paper maps and most two-dimensional digital maps reduce the three-dimensional world to a flat representation and therefore fail to convey terrain relief directly (Kim & Doeller, 2022). In actual tourism contexts, navigation involves not only route selection but also the continuous integration of spatial relationships, elevation changes, and environmental cues. Consequently, in topographically complex destinations, the necessity of mentally transforming 2D map information into 3D spatial structures renders navigation an inherently high-load spatial task (Guo et al., 2023; Kim & Doeller, 2024; Ou et al., 2025), thereby undermining the effectiveness of 2D maps for route planning and active navigation. This cognitive burden is particularly pronounced for tourists with lower spatial ability (Tiwari et al., 2022). In addition, traditional maps lack scene recognition capabilities and provide limited support for indoor environments (Alarifi et al., 2016; Panda et al., 2020). Their static and non-interactive nature often leads to difficulties in interpretation (Laurini et al., 2008), making it harder for users to quickly locate desired destinations (Ruginski et al., 2022).

Under these circumstances, Augmented Reality (AR) offers new possibilities for navigation. By integrating cameras, positioning systems, and sensors on mobile devices, AR maps overlay route guidance, point-of-interest descriptions, and interactive tasks onto the real-world environment (Kaźmierczak et al., 2021; Утробина & Попов, 2021). These displays adjust automatically to the user's head movements or device orientation (Tran et al., 2023), allowing virtual objects to be perceived as physical entities (Yahya & Dahanayake, 2021). This convergence provides a more intuitive navigation experience (Schickler et al., 2020). In terms of spatial cognition, accessing navigational information directly within the field of view minimizes the need for mental rotation and spatial reasoning, thereby reducing cognitive load (Wen et al., 2014; Zhang et al., 2021). This direct visualization enhances tourists' sense of immersion and engagement (Zakariya et al., 2023), while facilitating the acquisition of landmark and route knowledge (Qiu et al., 2024; Sasaki & Yamamoto, 2021; Shamsuddin & Din, 2016), leading to a more coherent understanding of spatial structure (Zhao et al., 2023). Overall, AR navigation outperforms traditional maps in complex environments in terms of usability, enjoyment, and overall performance (Munir et al., 2022). While applications of AR in navigation, tourism, and information visualization continue to expand and research on real-scene AR mapping technology advances steadily (Asraf et al., 2020; Geiger et al., 2014; Paucher & Turk, 2010), there remains a lack of research exploring user acceptance mechanisms. A small number of empirical studies have begun to explore AR acceptance in contexts such as shopping, cultural venues, and indoor campus navigation (Marian-Vladut et al., 2025; Said et al., 2023; Yang, 2023). However, these scenarios are typically situated in controlled environments with clear boundaries and singular tasks, where spatial complexity and real-time decision-making pressure are limited. In contrast, tourist attractions are often located in open outdoor settings characterized by complex topography and shifting visual references. Navigating these areas requires continuous path selection and spatial judgment; such high-task-oriented and cognitively demanding navigation may fundamentally shift user acceptance mechanisms. Furthermore, existing research predominantly relies on head-mounted AR devices, which differ fundamentally in interaction logic and usage patterns from mobile-based AR maps. Therefore, this study focuses on AR real-scene maps in tourist destinations and develops an user acceptance framework, aiming to elucidate the factors influencing behavioral intentions within complex and dynamic environments. This research not only addresses the gap in AR acceptance studies within complex outdoor contexts but also provides practical references for the design and promotion of digital guidance systems in scenic areas.

The structure of this article is as follows: The first section introduces the research background and problem statement. The second section reviews related studies, including TAM and AR map literature. The third section presents the research framework and key hypotheses. The fourth section describes the research materials, questionnaire design, and data analysis methods. The fifth section reports the results of model evaluation, path analysis, and hypothesis testing. The sixth section discusses the effects of various factors on AR map acceptance and the role of demographic differences. Finally, the study concludes with a summary of findings, limitations, and directions for future research.

2. Related research

2.1. Augmented reality (AR) maps

As a spatial visualization form integrating Geographic Information Systems (GIS), sensor technology, and mobile computing, AR maps achieve interactive spatial information presentation by blending virtual objects with real-world scenes (Ma et al., 2021; Werner, 2018). Their core value lies in the dynamic display of spatial information aligned with the user's perspective through multi-sensor fusion technology. This approach not only addresses positioning challenges in complex indoor and outdoor environments but also significantly enhances user spatial perception and navigation experiences through the real-time presentation of semantic information.

Early exploration of AR maps was predominantly centered within the fields of computer graphics and human-computer interaction (HCI). Research during this phase largely prioritized technical architecture and performance metrics, such as the optimization of positioning accuracy and text recognition capabilities (Huang et al., 2019; Lu et al., 2021). The explicit classification of Mixed Reality (MR) environments and the systematic identification of AR's technical challenges and advantages established the theoretical framework and academic foundation for subsequent map system development (Azuma, 1997; Milgram et al., 1995). AR navigation primarily emerged in the form of experimental projects conducted in laboratory settings, such as the "Touring Machine" system. By superimposing 3D virtual data in urban settings using HMDs, it demonstrated the nascent potential of AR for geographic navigation (Feiner et al., 1997). Nonetheless, due to limited hardware and processing capabilities, early AR maps largely relied on simple overlay approaches based on image recognition and visual markers (N. Zou et al., 2023). At the same time, early systems typically required the integration of multiple external sensors, resulting in high costs, bulky hardware, and limited portability. These constraints substantially restricted the practical application and large-scale adoption of AR maps. With the widespread adoption of smartphones, the development of AR maps entered a phase of rapid advancement. The integration of multiple sensors within mobile devices provided a low-cost hardware foundation for AR systems, enabling a transition from simple visual overlays to more advanced forms of environmental perception (Lu et al., 2024). During this stage, Simultaneous Localization and Mapping (SLAM) technology emerged as a cornerstone of AR mapping, enabling devices to concurrently achieve autonomous localization and map construction in unknown environments. This significantly bolstered system stability and spatial consistency (Carmigniani et al., 2011), facilitating high-precision positioning and dynamic environment modeling (Cheliotis et al., 2023; Theodorou et al., 2022). The integration of inertial sensors and dedicated chips for Visual-Inertial Odometry (VIO) has enabled mobile devices to achieve high-precision real-time positioning while maintaining low power consumption (Gerstweiler et al., 2015). On the application level, the maturation of mainstream mobile SDKs and third-party tools has significantly lowered development barriers, propelling AR maps from controlled laboratory environments into diverse real-world settings, such as shopping malls, museums, and urban streets (Gozali et al., 2024; Rohil & Ashok, 2022; Said et al., 2023; Yang, 2023).

Presently, AR maps have achieved significant breakthroughs in both underlying algorithms and hardware platforms, with smartphone-based AR maps emerging as the dominant application form. By incorporating advanced modeling techniques such as point cloud maps, AR maps deliver a more immersive and spatially intuitive navigation experience compared to traditional 2D maps (Qiu et al., 2024). Consequently, AR mapping is increasingly characterized by interdisciplinary development, integrating geospatial information with cognitive psychology to achieve the intelligent representation of geographic data (Amirian et al., 2015). The maturation of these technologies has directly facilitated the extensive application of AR maps in specific contexts like tourist attractions, establishing a new digital guidance paradigm characterized by the fusion of virtual and real environments.

In summary, AR maps have evolved from laboratory research to practical applications, with their core advantage lying in the integration of software and hardware sensing to transform static data into intuitive, virtual-real fused spatial information, thereby substantially enhancing the expressiveness of

geographic information. However, existing studies remain heavily focused on technical aspects such as algorithm optimization, while attention to user-related dimensions has lagged behind. Users are often treated as passive recipients, and research contexts are largely confined to stable environments, with limited investigation of outdoor settings. In particular, issues related to users' psychological perceptions, behavioral intentions, and underlying motivations remain underexplored, indicating that theoretical research on human–computer interaction and user experience in AR mapping still requires further development.

2.2. Application of augmented reality in tourism contexts

Augmented reality is a technology that integrates computer-generated virtual information with the real-world environment in real time, thereby enhancing user perception and interaction (Azuma, 1997). By superimposing digital content such as images, sounds, texts, and videos, it enriches the informational layer of the physical environment and enables users to access and understand information in a more intuitive and immersive way. AR technology originated in the 1960s (Sutherland, 1968), and with the improvement of mobile device performance and the development of sensing technologies, it has gradually evolved from laboratory research to commercial applications, demonstrating great potential in the tourism industry (Jalilvand & Ghasemi, 2024).

In the field of cultural heritage and attraction interpretation, AR technology provides tourists with immersive and interactive learning experiences. Jiang et al. (2023) found that AR can reconstruct the original appearance of historical sites, allowing visitors to reexperience historical scenes and strengthen memory and identification with cultural heritage. Chen et al. (2023) pointed out that interactive interpretation enabled by AR deepens tourists' understanding of cultural contexts and promotes the shift from passive reception to active exploration. Zhou et al. (2019) proposed an AR-based guide system that applies image recognition to scan the surrounding environment in real time, identify nearby attractions, and overlay customized virtual information through interactive interfaces, thereby improving the efficiency of interpretation.

In terms of interactive experiences, AR can effectively stimulate tourists' curiosity and sense of immersion, enhancing their emotional connection with destinations. Tyagi et al. (2022) indicated that AR technology increases the interactivity between tourists and attractions through games and interactive storytelling, making the travel experience more vivid and engaging. Jiman and Kulal (2023) demonstrated that AR provides visitors with new modes of interaction by superimposing digital information onto the physical world, simplifying access to information, improving convenience and efficiency, and enhancing tourism experiences across various aspects such as information acquisition, navigation, and cultural heritage appreciation. Some AR applications also allow users to mark and share points of interest, enhancing interactivity and social engagement during travel (Evagelou et al., 2024; Ghandour et al., 2021).

In hotel and accommodation services, AR technology provides users with functions such as virtual hotel tours, room information inquiries, interactive menu browsing, and personalized service recommendations. Gudadhe et al. (2021) pointed out that AR enables tourists to preview and book hotels at any time and place, obtain indoor navigation information, translate signage, and locate dining and entertainment options through interactive and intuitive interfaces, thereby transforming the accommodation experience. Kabadayı (2020) also showed that AR applications are widely adopted across various types of accommodation to enhance service quality in the hospitality industry. The study further indicated that these applications assist users in making decisions.

In transportation and navigation, AR technology supports tourists' travel through real-time information provision, route guidance, and interactive experiences (Zhuang et al., 2022). In marketing and promotion, AR technology provides innovative strategies for tourism destinations (Rahimi et al., 2017). In summary, AR technology, through new modes of information presentation, has reshaped the interaction between tourists and destinations. It not only enhances engagement and optimizes experience processes but also drives the upgrading of tourism services and experiences (Aggag & Kortam, 2025; Maria et al., 2024).

2.3. Application of maps in tourism contexts

Maps have always played an indispensable role in tourism. de Oliveira (2005) and Eboy (2017) pointed out that maps are not only fundamental media for route selection, destination marketing, and spatial cognition, but also serve as irreplaceable tools in tourism planning, information communication, and cultural expression by conveying spatial information and integrating tourism resources. With the advancement of the internet and mobile communication technologies, maps have evolved from static paper-based forms into dynamic and interactive digital systems (Dickmann, 2005). Modern mobile navigation technologies are evolving toward greater intelligence and personalization (Cheverst et al., 2000; Çilkin & Toksöz, 2024). Current research on the application of digital maps in tourism contexts can be categorized into three areas: pre-trip planning, on-site navigation, and cultural communication.

Pre-trip planning aims to provide comprehensive information support for travel decision-making and itinerary design (Pabel & Prideaux, 2016). Tourists often use digital maps and mobile applications, such as Google Maps and Apple Maps, to understand the layout of destinations, transportation routes, and the distribution of attractions, while also comparing multiple routes and accessing booking services (Chai-Arualert et al., 2023; Gupta & Dogra, 2017). In addition, web-based GIS planning and sharing systems allow tourists to select attractions through digital maps and share their plans with others (Koga & Yamamoto, 2021). Such systems improve the collaboration and flexibility of pre-trip preparation (Guerra & Boutoura, 2001).

During the trip, the function of digital maps extends from static planning to dynamic navigation and real-time information acquisition. Kostin (2018) highlighted that the integration of GPS positioning and augmented reality significantly improves navigation accuracy and interactivity. Karaş et al. (2021) confirmed that digital map route planning systems can recommend routes and points of interest based on users' real-time locations, thereby reducing disorientation and time loss. Bashir et al. (2024) provided tourists with real-time information and navigation services through mobile applications and interactive maps. Norrie and Signer (2005) studied digitally enhanced maps that offered interactive services through digital overlays, which further enrich the tourist experience.

Beyond navigation functions, digital maps also have significant value in cultural communication. Moss (2015), through multi-regional case studies, found that tourism maps could strengthen regional cultural identity and support destination marketing. Harede and Fekry Ibrahim (2019) created interactive digital tourism maps using Google Maps and geographic database technologies, which promoted tourism resource dissemination and cultural communication. Pál and Albert (2019) verified the positive role of digital maps in visitor education and guidance within geoparks.

Evidently, from paper maps to mobile digital navigation systems, mapping technology has become fully embedded in every stage of the tourism experience, serving as a vital link between tourists and their destinations. With the deep integration of AR and tourism mapping, the map is evolving from a traditional navigational tool into an immersive carrier that fuses virtuality with reality. This evolution demonstrates unique value in enhancing navigational precision, strengthening cultural dissemination, and facilitating the provision of personalized services.

2.4. User acceptance of augmented reality

Research on user acceptance of Augmented Reality has been explored across various scenarios, with the majority of studies employing the Technology Acceptance Model (TAM) as the core theoretical foundation or extending it for specialized applications.

In cultural heritage museums, Xu et al. (2025) examined an AR cultural interpretation system at the Sanxingdui archaeological site based on an extended TAM. Their study found that personal innovativeness and sense of security have a positive impact on attitudes toward AR use. Furthermore, as the core variables of the model, perceived ease of use and perceived usefulness remain the key factors in shaping users' positive attitudes and subsequent behavioral intentions. In the field of education and training, Papakostas et al. (2021) employed an extended TAM to evaluate user acceptance of the Naval Fire Fighting Training & Education System (NAFTES), a

mobile augmented reality training system designed for firefighters. Their results indicate that usability, perceived ease of use, and perceived interactivity are key prerequisites influencing user decision-making, with system usability identified as the strongest predictor of usage intention. In the domain of tourism mobile applications, Omran et al. (2025) investigated VR/AR-based tourism mobile apps by integrating text mining with partial least squares structural equation modeling (PLS-SEM) to systematically examine how application attributes influence tourists' behavioral engagement. The results indicate that perceived usefulness, perceived ease of use, and immersive characteristics exert significant positive effects on user behavioral engagement through app satisfaction, with immersive experience and perceived enjoyment playing a critical role in shaping user attitudes. In consumer application contexts, Lai et al. (2025) integrated the Technology Acceptance Model (TAM) with the Value-based Adoption Model (VAM) to examine the formation of users' intention to adopt augmented reality in online shopping. Using data from Generation Z consumers and applying partial least squares structural equation modeling (PLS-SEM), the study found that perceived usefulness, perceived ease of use, and immersive experience significantly enhance user engagement and adoption intention through perceived value. Álvarez-Marín et al. (2025) incorporated esthetics and information quality into the Technology Acceptance Model (TAM) to examine the mechanisms underlying users' adoption of augmented reality (AR) technology. The findings indicate that the visual esthetics of AR systems influence users' perceived usefulness, while high information quality significantly strengthens the relationship between perceived usefulness and adoption intention. Overall, these studies primarily examine user acceptance from three perspectives: technical performance, individual user characteristics, and the psychological experience of interaction.

Although user acceptance of AR maps has received increasing attention, existing research remains limited. Research by He and Wang (2025), grounded in the Stimulus-Organism-Response (SOR) model, explored the factors influencing tourists' adoption of AR museum guidance services. Their findings reveal that information quality and service quality are critical to enhancing AR acceptance. Specifically, personalization, interactivity, authenticity, and esthetics serve as external stimuli that collectively drive behavioral intentions (Response) by eliciting users' perceived pleasure and satisfaction (Organism). Said et al. (2023) used the UTAUT model to study indoor AR navigation among Singaporean consumers. The results showed that performance expectancy and facilitating conditions are the strongest predictors of behavioral intention. Additionally, perceived ease of use and perceived usefulness remain core factors for accepting AR spatial services. Based on the General Extended Technology Acceptance Model for E-Learning (GETAMEL), Y. Liu et al. (2023) compared AR and 3D mobile guidance systems developed by the National Taiwan University Library. The study found that subjective norms, hedonic experience, perceived usefulness, and perceived ease of use are key drivers of AR service acceptance, with sensory pleasure further enhancing adoption motivation. These findings suggest that, in map-based navigation contexts, factors influencing user acceptance extend beyond functional attributes such as system accuracy and ease of use to include affective factors, such as the enjoyment and pleasure users experience during navigation.

However, the acceptance mechanisms of AR real-scene maps exhibit notable context-specific characteristics in particular environments. While existing studies provide empirical support for understanding AR map acceptance, their findings primarily rely on application scenarios with stable information and environments. For expansive outdoor scenic areas with complex terrain and dynamic environments, AR maps face greater challenges in positioning accuracy, information presentation, and interaction experience. These factors impose more stringent requirements on system and information quality (Brata et al., 2024; Ma et al., 2021). More importantly, existing research on the acceptance of AR technology and AR real-scene maps tends to focus on single functions, such as navigation or interpretation, while lacking a systematic investigation into the integration of these features. In such an integrated model, users seek both the instrumental value of navigation and the hedonic, immersive experiences derived from informational interpretation and social interaction. However, there is still a lack of in-depth research on how this functional complexity alters user perception and acceptance mechanisms.

In summary, existing research has confirmed the significance of classic variables such as perceived usefulness, while highlighting the critical roles of system information quality and interactive enjoyment in AR acceptance. However, in complex and dynamic environments such as outdoor tourist destinations, the mechanisms underlying users' behavioral intentions become more intricate. Therefore, it is necessary to further examine how technological performance and affective value jointly influence tourists' decision-making on the basis of traditional models, in order to achieve a deeper understanding of AR map acceptance in complex contexts.

3. Proposed model and hypotheses

As a cultural tourism application that integrates real-world information with digital navigation, the use of AR maps is influenced by multiple factors, including system quality, information quality, immersion, and perceived enjoyment. System and information quality determine the reliability and accuracy of technical services (DeLone & McLean, 2003), while immersion and perceived enjoyment shape users' sensory interaction and emotional engagement in virtual-real integrated contexts (Leveau & Camus, 2023). These factors collectively constitute key variables influencing users' technology adoption. Given that AR maps are primarily used by general visitors and often applied in time-constrained, task-oriented tourism contexts, users' adoption decisions largely depend on whether the system can be quickly understood and effectively support core functions such as navigation and information acquisition. The TAM, with its theoretical framework centered on perceived usefulness and perceived ease of use, aligns perfectly with the decision-making foundations of such functional-oriented applications. Furthermore, the model's high extensibility makes it suitable for integrating system characteristics and experiential factors. Therefore, this study adopts system quality, information quality, immersion, and perceived enjoyment as independent variables to explore how they influence users' behavioral intentions through the mediation of perceived usefulness and perceived ease of use (Figure 1).

3.1. System quality

System quality (SQ) is a key dimension for assessing the overall performance of information systems, encompassing reliability, flexibility, ease of use, and other aspects (Nelson et al., 2005). Multiple studies have confirmed that system quality has a significant positive impact on perceived ease of use and perceived usefulness (Davis, 1989; İnci, 2017; Rana et al., 2015). Moreover, system quality also affects user satisfaction, behavioral intention, continued usage, and recommendation intention (Jung et al., 2015; Zhou & Wang, 2016). For AR applications, Cao et al. (2023) indicated that system quality directly influences users' perception of realism and interaction fluency, highlighting that precise object tracking, low latency, and stable rendering are critical for enhancing user satisfaction with AR experiences. Gao et al. (2023) and Wang et al. (2021) found that system quality significantly affects users' route planning

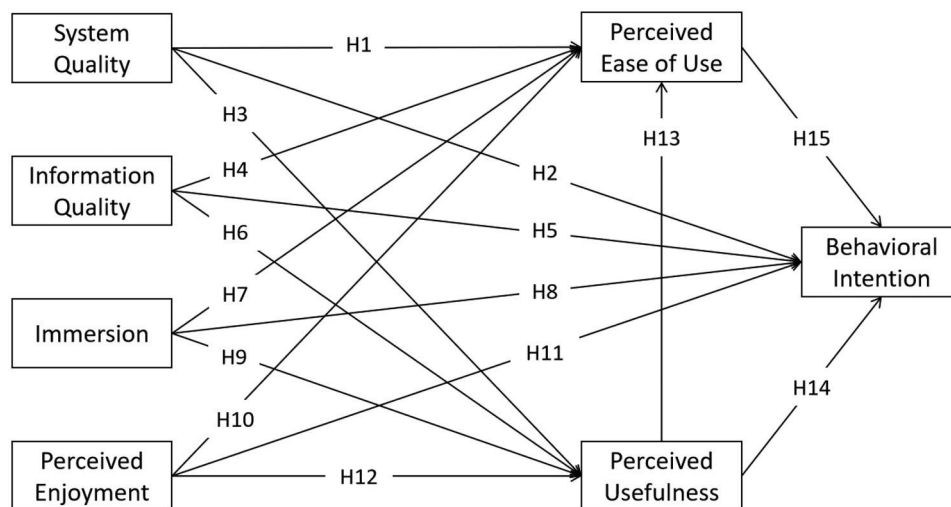


Figure 1. Research model for measuring the adoption of tourists' intention to use AR real-scene maps.

efficiency by influencing navigation accuracy and map data precision. This underscores the key role of system quality in optimizing navigation reliability and supporting efficient wayfinding. Therefore, this study proposes the following hypotheses:

H1: System quality has a positive effect on perceived usefulness.

H2: System quality has a positive effect on behavioral intention.

H3: System quality has a positive effect on perceived ease of use.

3.2. Information quality

Information quality (IQ) refers to the desirable characteristics of system outputs, such as relevance, accuracy, timeliness of comprehension, completeness, usability, and accessibility (DeLone & McLean, 2003). As one of the core competencies of AR map systems, high-quality information is typically characterized by accuracy, relevance, timeliness, and ease of understanding. These factors can not only significantly enhance users' perceived usefulness and perceived ease of use, but also further strengthen their behavioral intention for continued use. Gupta and Sharma (2024) found that when the information provided by the system does not match users' interests, users tend to perceive the system as lacking value. Huynh et al. (2019) confirmed that content professionalism and reliability help enhance users' trust and increase their reliance on system functionalities. Specifically, in AR navigation applications, high-quality information can assist users in accurately locating target destinations and selecting efficient routes, thereby improving overall navigation efficiency and enhancing perceptions of usefulness and ease of use. Therefore, the following hypotheses are proposed:

H4: Information quality has a positive effect on perceived usefulness.

H5: Information quality has a positive effect on behavioral intention.

H6: Information quality has a positive effect on perceived ease of use.

3.3. Immersion

Immersion (IM) refers to a state in which individuals are fully engaged in an activity, feeling present in the environment and unaware of distractions (Witmer & Singer, 1998). In AR applications, immersion is achieved through the overlay of virtual information onto real-world scenes, enhancing users' perception and interaction with the environment and creating a unique immersive experience. As a psychological state of deep engagement in interactive technologies or media, immersion has been widely explored in technology acceptance and user behavior research in recent years. It not only enhances interactive experience but also directly influences users' attitudes toward acceptance and behavioral intentions (Zhang et al., 2020). Tang et al. (2022) found that higher immersion in fitness applications improves users' perceived usefulness. Nhan et al. (2022) reported that immersive experiences drive users' emotions, values, and behaviors in mobile AR apps. Yen and Dong (2025) confirmed that immersion significantly influences satisfaction and behavioral intention in mobile games. Based on these insights, this study treats immersion as a key external variable affecting AR experience and proposes:

H7: Immersion has a positive effect on perceived usefulness.

H8: Immersion has a positive effect on behavioral intention.

H9: Immersion has a positive effect on perceived ease of use.

3.4. Perceived enjoyment

Perceived enjoyment (EN) is commonly defined as the pleasure and satisfaction individuals experience through sensory and cognitive engagement when using information systems, participating in activities,

or interacting with products. The concept was first proposed by Davis (1989). It was used to explain user acceptance of computer systems, highlighting that perceived enjoyment is a key factor directly influencing usage intention and satisfaction (Davis et al., 1989). As an intrinsic motivator, perceived enjoyment affects perceived ease of use and perceived usefulness both directly and indirectly, and strongly influences behavioral intention. Studies have shown that perceived enjoyment significantly predicts usage intention (Putri et al., 2025), and in AR applications, it drives perceived usefulness and purchase intention. Based on these insights, the following hypotheses are proposed:

H10: Perceived enjoyment has a positive effect on perceived usefulness.

H11: Perceived enjoyment has a positive effect on behavioral intention.

H12: Perceived enjoyment has a positive effect on perceived ease of use.

3.5. Perceived usefulness and perceived ease of use

User experience is a central dimension for exploring user acceptance of new technologies. According to the user experience model proposed by Hassenzahl (2018), the pragmatic dimension is the primary standard for individual evaluation. It focuses on whether a product possesses the functions to support task completion and the convenience of its operation. In this study, this dimension is represented by perceived usefulness and perceived ease of use within the TAM. The TAM model indicates that perceived usefulness (PU) refers to users' belief that using a technology can enhance task performance, while perceived ease of use (PE) reflects the degree of mental effort felt by users during the interaction process (Davis, 1989). Previous studies have shown that PE not only directly affects behavioral intention (BI) but also indirectly enhances user acceptance by increasing perceived usefulness (Venkatesh & Davis, 2000). In AR navigation scenarios, perceived usefulness reflects whether users believe the AR map can improve spatial positioning and route judgment, whereas perceived ease of use reflects the ease of operating the AR interface and recognizing virtual information. Greater operational simplicity reduces cognitive load, allowing users to focus more on navigation tasks, thereby strengthening perceptions of the system's functional value and ultimately increasing behavioral intention. Based on these insights, the following hypotheses are proposed:

H13: Perceived ease of use has a positive effect on perceived usefulness.

H14: Perceived ease of use has a positive effect on behavioral intention.

H15: Perceived usefulness has a positive effect on behavioral intention.

4. Methodology

4.1. Materials

This study selected the AR map of Detian Transnational Waterfall Scenic Area as the research subject. The Detian Transnational Waterfall Scenic Area is located in Chongzuo, Guangxi, China. It is a 5 A-level tourist attraction in China, the largest transnational waterfall in Asia, and the fourth largest in the world, spanning the border between China and Vietnam. Influenced by karst topography, the area forms a multi-level spatial layout, and attractions such as viewing platforms, cross-border bamboo rafts, and high-altitude glass walkways generate complex movement patterns, presenting visitors with significant navigational and orientation challenges. In 2024, the scenic area officially launched the "Detian Guide" mini-program. The program overlays virtual tour guides, scenic spot descriptions, route arrows, and distance labels onto real-world scenes through smartphone cameras, providing real-time navigation and information delivery. Users can also engage in activities within the scenic area through features such as favoriting, check-ins, sharing, and interactive tasks. Operating within the WeChat ecosystem, the system requires no additional installation, thereby lowering user operational barriers. This scenic area attracts a large and diverse group of visitors, covering a wide range of ages and technological proficiency levels, which allows for a comprehensive examination of user intentions in complex environments.



Figure 2. Interface of the “detian guide” mini-program: (a) main interface, (b) destination selection interface, (c) scenic spot information interface, and (d) navigation interface.

Overall, the program represents a mature implementation of mobile AR navigation in tourism contexts, ensuring its stability and reliability as research material. The main interface of the “Detian Guide” mini-program, along with sample scenic spot markers, information displays, and AR navigation paths, is shown in Figure 2.

4.2. Questionnaire development

The questionnaire consisted of two sections. The first section collected demographic information, including gender, age, and prior experience with AR applications. The second section focused on users’ perceptions and evaluations of the “Detian Guide” mini-program, encompassing seven dimensions, namely system quality, information quality, immersion, perceived enjoyment, perceived ease of use, perceived usefulness, and behavioral intention.

All measurement items were adapted from validated scales and tailored to the specific context of AR navigation. Specifically, items for system quality and immersion were adapted from Cheng et al. (2024); information quality and perceived enjoyment items were adapted from Williams et al. (1992) and Huang et al. (2013); perceived ease of use, perceived usefulness, and behavioral intention were constructed according to the classical definitions of the Technology Acceptance Model (Davis, 1989). The final questionnaire included four items for information quality and three items for each of the remaining dimensions, meeting the minimum requirements for structural equation modeling (Byrne, 2001).

To minimize measurement error due to respondent fatigue, all items were phrased concisely and clearly, and responses were recorded using a 5-point Likert scale (1 = “strongly disagree,” 5 = “strongly agree”). Prior to formal administration, a small-scale pilot test was conducted with 30 randomly selected tourists. The results indicated that most participants found the items easy to understand, and the average completion time ranged from 3 to 5 min. Based on feedback from four participants who suggested including a privacy statement, a note was added at the beginning of the questionnaire clarifying that “data will be used solely for academic research and will remain strictly anonymous.” The final questionnaire items are presented in Table 1.

4.3. Survey design

The field survey was conducted within the Detian Transnational Waterfall Scenic Area, focusing on areas with dense visitor presence and high demand for AR navigation, such as around the waterfall

Table 1. Measurement items and corresponding reference.

Construct	Item	Measurement	Reference
System quality	SQ1	The AR navigation system runs smoothly with little lag or crashes.	Cheng et al. (2024)
	SQ2	The system interface is clear and navigation is easy to use.	
	SQ3	The AR interface design is attractive, with images and animations enhancing enjoyment.	
Information quality	IQ1	The AR system provides accurate and reliable information matching the real scene.	Williams et al. (1992)
	IQ2	The system shows complete navigation details including directions and routes.	
	IQ3	The information is presented clearly and concisely, using easy-to-understand language without causing confusion.	
	IQ4	The information provided by the system helps me quickly locate target destinations.	
Immersion	IM1	Using the mobile AR navigation helps me focus on and understand my surroundings.	Cheng et al. (2023)
	IM2	The AR navigation system immerses me in the tour experience and enhances my sense of presence.	
	IM3	I am not easily distracted by other things while using the system.	
Perceived enjoyment	EN1	Using the AR navigation is an enjoyable experience.	Huang et al. (2013)
	EN2	I feel pleasure during the use of the system.	
	EN3	The system makes my visit more fun and engaging.	
Perceived ease of use	PE1	Learning to use the AR navigation system is easy for me.	Davis (1989)
	PE2	I quickly mastered how to operate the AR navigation.	
	PE3	The AR navigation helps me complete navigation tasks effortlessly.	
Perceived usefulness	PU1	This AR navigation system has shortened the time it takes me to find routes.	Davis (1989)
	PU2	Using this system helps me better understand the layout of the scenic area.	
	PU3	The AR navigation system is very helpful for my overall visiting experience.	
Behavioral intention	BI1	I would use this AR navigation system again in the future.	Davis (1989)
	BI2	If other scenic areas offer AR navigation, I would be willing to try it.	
	BI3	I would recommend this AR navigation experience to others.	

viewing platforms and near Scenic Marker 53. These locations offered good signal coverage and open conditions, making them suitable for participants to experience the AR map and complete assigned tasks.

Participants were recruited on-site. Researchers invited visitors who were using or willing to try the “Detian Guide” mini-program and clearly explained the study’s purpose, task procedures, and data usage to ensure informed and voluntary participation. After signing informed consent forms, participants completed an AR navigation task under the guidance of the researchers, which took approximately 5–8 min. The task involved opening the mini-program, performing location and destination selection, and navigating along real-world paths to the target plot. Upon completion of the navigation task, participants filled out a questionnaire, which took approximately 3–5 min. Researchers supervised the process and collected the questionnaires on-site. All data were anonymized and used exclusively for academic research. Each valid participant received a souvenir featuring the characteristics of the scenic area as a token of appreciation.

4.4. Participants

A total of 400 paper-based questionnaires were distributed. Prior to formal analysis, rigorous data screening was conducted to ensure quality. Incomplete questionnaires ($n=21$) and those with evident logical inconsistencies (e.g., contradictory responses to related items, $n=67$) were excluded. The final sample consisted of 322 valid responses, yielding an effective response rate of 80%.

Regarding demographic characteristics, the sample included 154 males (47.8%) and 168 females (52.2%). The 18–24 age group accounted for the largest proportion (41.9%), followed by 25–34 age group (34.5%), with the remaining participants distributed across other age brackets. Concerning prior

Table 2. Demographic characteristics of participants ($n = 322$).

Variable	Category	Frequency	Percentage (%)
Gender	Male	154	47.8
	Female	168	52.2
Age	18–24 Years	135	41.9
	25–34 Years	111	34.5
	35–44 Years	44	13.7
	>46 Years	32	9.9
Experience	Yes	123	38.2
	No	199	61.8

Table 3. Structural reliability and convergent validity.

Construct	Item	Cronbach's α	Standardized loading	CR	AVE
System quality	SQ1	0.790	0.715	0.853	0.541
	SQ2		0.759		
	SQ3		0.766		
Information quality	IQ1	0.847	0.738	0.896	0.571
	IQ2		0.783		
	IQ3		0.784		
	IQ4		0.748		
Perceived immersion	IM1	0.797	0.718	0.847	0.535
	IM2		0.734		
	IM3		0.814		
Perceived enjoyment	EN1	0.827	0.815	0.877	0.591
	EN2		0.735		
	EN3		0.802		
Perceived ease of use	PE1	0.823	0.751	0.848	0.521
	PE2		0.688		
	PE3		0.757		
Perceived usefulness	PU1	0.804	0.741	0.852	0.537
	PU2		0.751		
	PU3		0.768		
Behavioral intention	BI1	0.793	0.741	0.845	0.525
	BI2		0.7		
	BI3		0.777		

experience with AR maps, 123 respondents (38.2%) reported having relevant experience, while 199 (61.8%) were first-time users. Overall, the sample composition closely reflects the demographic profile of visitors to the Detian Transnational Waterfall Scenic Area (Table 2).

4.5. Data analysis

This study employed SPSS 26 and AMOS 23 for data analysis. Reliability and exploratory factor analyses were first conducted in SPSS to assess the internal consistency and dimensional validity of the scales. Confirmatory factor analysis was then performed in AMOS to evaluate structural validity. Descriptive statistics and correlation analyses were subsequently carried out in SPSS to examine the distribution characteristics of the variables and their interrelationships. Structural equation modeling (SEM) was used to test the hypothesized paths. Finally, hierarchical regression analyses were employed to test the moderating effects of gender, age, and prior experience on the core paths of the model.

5. Results

5.1. Measurement model

This study assessed the reliability and convergent validity of each latent variable using Cronbach's α , Composite Reliability (CR), and Average Variance Extracted (AVE). A Cronbach's α greater than 0.7 and a CR above 0.7 indicate good internal consistency and reliability (Hair et al., 2012), while an AVE exceeding 0.5 suggests satisfactory convergent validity (Fornell & Larcker, 1981). Standardized factor loadings of observed variables on their respective latent constructs were also considered, with loadings above 0.5 generally indicating that items adequately represent their constructs. As shown in Table 3, the CR values for all seven latent variables exceeded 0.8, the AVE values were above 0.5, and all item loadings were greater than 0.6. These results collectively indicate that all constructs demonstrate good

Table 4. Factor loadings after varimax rotation.

Item	Component						
	1	2	3	4	5	6	7
SQ1							0.751
SQ2							0.776
SQ3							0.769
IQ1	0.701						
IQ2	0.806						
IQ3	0.792						
IQ4	0.757						
IM1				0.773			
IM2				0.782			
IM3				0.826			
EN1		0.822					
EN2		0.818					
EN3		0.838					
PE1			0.774				
PE2			0.835				
PE3			0.799				
PU1					0.823		
PU2					0.77		
PU3					0.782		
BI1						0.8	
BI2						0.785	
BI3						0.801	

Table 5. Descriptive statistics and normality test results.

	Mean	Standard deviation	Skewness	Kurtosis
SQ1	3.404	1.140	-0.319	-0.648
SQ2	3.394	1.206	-0.270	-0.814
SQ3	3.379	1.210	-0.335	-0.669
IQ1	3.500	1.185	-0.402	-0.654
IQ2	3.525	1.210	-0.329	-0.822
IQ3	3.419	1.195	-0.381	-0.679
IQ4	3.447	1.113	-0.228	-0.555
IM1	3.419	1.254	-0.346	-0.885
IM2	3.304	1.158	-0.262	-0.645
IM3	3.407	1.149	-0.347	-0.608
EN1	3.227	1.278	-0.116	-0.997
EN2	3.071	1.255	-0.069	-0.862
EN3	3.134	1.245	-0.158	-0.841
PE1	3.450	1.194	-0.329	-0.763
PE2	3.394	1.198	-0.306	-0.715
PE3	3.404	1.157	-0.311	-0.686
PU1	3.348	1.151	-0.352	-0.565
PU2	3.481	1.174	-0.286	-0.660
PU3	3.419	1.131	-0.254	-0.690
BI1	3.286	1.184	-0.274	-0.759
BI2	3.301	1.173	-0.185	-0.645
BI3	3.258	1.116	-0.089	-0.544

convergent validity and composite reliability, supporting their use in subsequent structural model analyses.

The data were analyzed using the varimax rotation method to clarify the correspondence between factors and measurement items. As shown in Table 4, in the rotated component matrix, all communalities exceeded 0.7, jointly explaining 72.21% of the total variance. This indicates a strong association between items and factors, suggesting that the factors effectively capture the underlying constructs. Moreover, the grouping of measurement indicators under each variable was consistent with theoretical expectations.

Normality of the measurement items was assessed using skewness and kurtosis. Following the criteria proposed by Kline (2023), data can be considered approximately normally distributed when the absolute value of skewness is less than 3 and the absolute value of kurtosis is less than 8. As shown in Table 5, all skewness and kurtosis values for the measurement items in this study fall within the recommended thresholds, indicating that the data satisfy approximate normality.

Table 6. Results of correlation analysis matrix for variables.

Variable	SQ	IQ	IM	EN	PE	PU	BI
SQ	1						
IQ	0.476**	1					
IM	0.452**	0.353**	1				
EN	0.362**	0.353**	0.268**	1			
PE	0.338**	0.393**	0.369**	0.223**	1		
PU	0.401**	0.431**	0.294**	0.329**	0.396**	1	
BI	0.291**	0.392**	0.319**	0.262**	0.333**	0.400**	1

Note: ** indicates significance at the 0.01 level (two-tailed).

Table 7. Model fitting index results.

Indicator	χ^2/df	GFI	CFI	NFI	RMSEA
Judgment Criteria	1–3	>0.8	>0.8	>0.8	<0.08
Value	2.417	0.871	0.904	0.848	0.066

5.2. Correlation analysis

To further examine the relationships among the variables, Pearson correlation analysis was conducted for all latent constructs, and the results are presented in Table 6. The correlation matrix indicates that all variables are significantly and positively correlated ($p < 0.01$). This finding suggests good inter-construct associations, consistent with theoretical expectations, and provides a solid foundation for subsequent SEM analysis.

5.3. Evaluation of structural model fit

To assess the goodness-of-fit of the structural model, a set of widely adopted fit indices was employed. Following the recommendations of Hair et al. (1998) such as GFI, CFI, and NFI exceed 0.80, and RMSEA is below 0.08. In this study, the structural model yielded $\chi^2/df = 2.417$, $GFI = 0.871$, $CFI = 0.904$, $NFI = 0.848$, and $RMSEA = 0.066$, all of which meet the aforementioned thresholds (Table 7). These results indicate that the proposed structural model demonstrates a satisfactory fit with the observed data, providing a robust statistical foundation for subsequent path analysis.

5.4. Path analysis and hypothesis testing

SEM was employed to examine the relationships among constructs and to test the proposed hypotheses. As shown in Table 8 and Figure 3, all hypotheses were supported except for H2, H7, H11 and H12. Specifically, SQ had a significant positive effect on PU and PE ($\beta = 0.126$, $p < 0.05$; $\beta = 0.186$, $p < 0.01$). IQ also positively influenced PU, BI and PE ($\beta = 0.264$, $p < 0.001$; $\beta = 0.224$, $p < 0.01$; $\beta = 0.326$, $p < 0.001$). IM did not significantly affect PU, but had a positive impact on BI and PE ($\beta = 0.152$, $p < 0.05$; $\beta = 0.205$, $p < 0.001$). EN significantly affected PU ($\beta = 0.115$, $p < 0.05$), while its effect on BI and PE was not significant. PE positively influenced PU and BI ($\beta = 0.236$, $p < 0.01$; $\beta = 0.177$, $p < 0.05$), and PU had positive effects on BI ($\beta = 0.187$, $p < 0.05$).

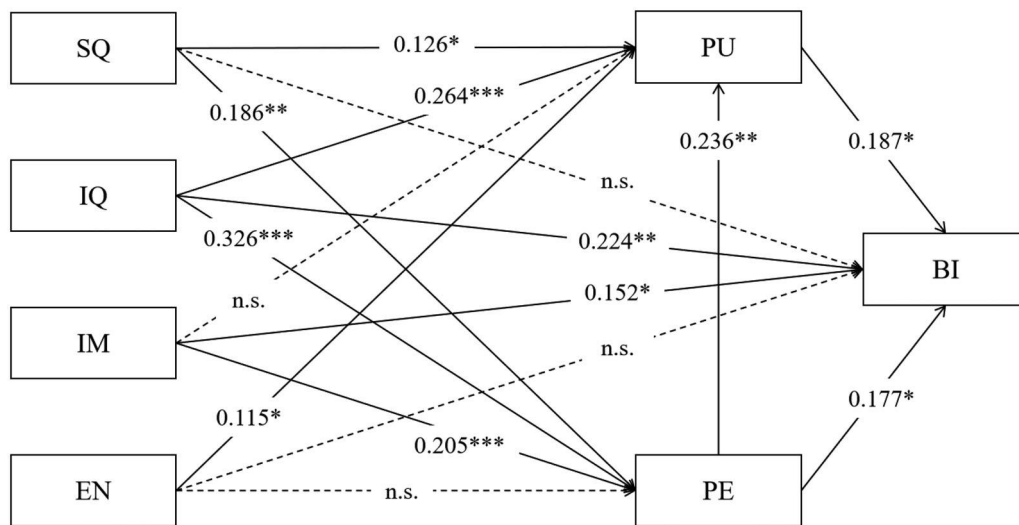
5.5. Analysis of demographic differences

To examine the moderating effects of gender, age, and prior experience on the core paths, this study employed hierarchical regression analysis. Continuous variables were mean-centered to mitigate multicollinearity, and moderation was determined based on the significance of the interaction term coefficients ($p < 0.05$). The results in Table 9 indicate that gender significantly and positively moderates the paths from $IQ \rightarrow PE$, $IM \rightarrow PE$, and $PE \rightarrow PU$, with all interaction coefficients being significantly positive and model explanatory power showing significant improvement. The moderating effect of age is relatively limited, occurring only as a negative moderation in the path from $PE \rightarrow BI$, suggesting that the influence of ease of use on behavioral intention diminishes with advancing age. Prior experience exhibits the most prominent and extensive moderating effect: in the paths of $SQ \rightarrow PU$, $IQ \rightarrow BI$, $PU \rightarrow BI$, and $PE \rightarrow PU$, the interaction terms all reached significance levels with marked increases in explanatory power. This demonstrates

Table 8. Hypothesis test result.

Hypothesis	Path	Standardized estimate (β)	S.E.	C.R.	Results
H1	SQ $\rightarrow$ PU	0.126*	0.061	2.076	Accepted
H2	SQ $\rightarrow$ BI	-0.003	0.062	-0.048	Rejected
H3	SQ $\rightarrow$ PE	0.186**	0.062	3.016	Accepted
H4	IQ $\rightarrow$ PU	0.264***	0.07	3.745	Accepted
H5	IQ $\rightarrow$ BI	0.224**	0.075	2.994	Accepted
H6	IQ $\rightarrow$ PE	0.326***	0.069	4.746	Accepted
H7	IM $\rightarrow$ PU	0.099	0.059	1.664	Rejected
H8	IM $\rightarrow$ BI	0.152*	0.062	2.46	Accepted
H9	IM $\rightarrow$ PE	0.205***	0.06	3.397	Accepted
H10	EN $\rightarrow$ PU	0.115*	0.053	2.156	Accepted
H11	EN $\rightarrow$ BI	0.078	0.055	1.411	Rejected
H12	EN $\rightarrow$ PE	0.078	0.055	1.418	Rejected
H13	PE $\rightarrow$ PU	0.236**	0.075	3.147	Accepted
H14	PE $\rightarrow$ BI	0.177*	0.079	2.248	Accepted
H15	PU $\rightarrow$ BI	0.187*	0.082	2.277	Accepted

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

**Figure 3.** Results of model and hypothesis testing.

that the aforementioned relationships are significantly stronger among inexperienced users than among experienced users (Experience coding: 1 = Experienced, 2 = Inexperienced).

6. Discussion

This study aimed to identify the key factors influencing tourists' use of AR real-scene maps and to examine the mechanisms through which these factors, along with demographic characteristics, affect behavioral intentions, using an extended TAM framework. Results from structural equation modeling indicate that the primary paths in the original TAM were supported, with both perceived usefulness and perceived ease of use exerting significant effects on behavioral intention. Among the added external variables, 11 out of 15 hypothesized relationships were supported, while hypotheses H2, H7, H11, and H12 were not confirmed. Analysis of demographic factors revealed that gender, age, and prior AR experience exerted differential effects on certain variables, providing additional insights into user preferences across different groups. Overall, the proposed model demonstrates strong explanatory power, effectively capturing the mechanisms underlying AR map acceptance in the context of cultural tourism.

6.1. System quality

The results indicate that system quality is positively associated with perceived usefulness and perceived ease of use, which in turn indirectly influence behavioral intention. However, the hypothesized direct effect of system quality on behavioral intention was not supported.

Table 9. Moderating effects of demographic variables.

Moderator	Path	Variables	Regression coefficients			Model fit indices		
			β	SE	t	R^2	ΔR^2	F
Gender	IQ $\rightarrow$ PE	IQ	0.409***	0.055	7.366	0.168	0.01	3.98
		Gender	0.094	0.107	0.883			
		IQ $\times$ Gender	0.223*	0.112	1.994			
	IM $\rightarrow$ PE	IM	0.383***	0.053	7.187	0.149	0.012	4.63
		Gender	-0.046	0.107	-0.43			
		IM $\times$ Gender	0.23*	0.107	2.151			
	PE $\rightarrow$ PU	PE	0.359***	0.047	7.589	0.176	0.016	6.07
		Gender	-0.117	0.097	-1.198			
		PE $\times$ Gender	0.234*	0.095	2.463			
Age	PE $\rightarrow$ BI	PE	0.312***	0.05	6.281	0.123	0.012	4.18
		Age	-0.007	0.053	-0.135			
		PE $\times$ Age	-0.113*	0.055	-2.045			
Experience	SQ $\rightarrow$ PU	SQ	0.379***	0.049	7.76	0.184	0.01	3.91
		Experience	-0.226*	0.1	-2.256			
		SQ $\times$ Experience	0.199*	0.1	1.977			
		IQ	0.374***	0.051	7.288	0.177	0.012	4.65
	IQ $\rightarrow$ BI	Experience	-0.228*	0.103	-2.221			
		IQ $\times$ Experience	0.231*	0.107	2.156			
	PU $\rightarrow$ BI	PU	0.368***	0.052	7.09	0.203	0.034	13.62
		Experience	-0.224*	0.102	-2.202			
		PU $\times$ Experience	0.407***	0.11	3.69			
	PE $\rightarrow$ PU	PE	0.346***	0.049	7.017	0.174	0.016	6.17
		Experience	-0.114	0.105	-1.085			
		PE $\times$ Experience	0.258*	0.104	2.484			

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Specifically, system quality exerts significant positive effects on both perceived usefulness and perceived ease of use. As a critical factor ensuring the smooth operation of a platform, system quality encompasses reliability, security, user interface design, response speed, and adaptability (Calisir et al., 2014; Lee & Sung, 2023; Zhou et al., 2022). Fitria et al. (2024) demonstrated that high system quality enhances user trust and acceptance of technological products, thereby increasing perceived usefulness. A well-designed system with intuitive layout, prompt response to commands, and operation logic aligned with user habits allows users to interact smoothly without extensive learning, reducing perceived difficulty and enhancing ease of use. Conversely, system lag, slow response, or positioning errors can generate operational frustration, amplifying perceptions of difficulty and diminishing user experience. The efficiency and reliability of a system form the foundation of user experience and are among the key factors in attracting and retaining users (Jo & Bang, 2023). High service quality helps establish the platform's reputation, enhance user loyalty, and increase behavioral intention (Ryu et al., 2014). In this study, the effect of system quality on perceived usefulness ($p < 0.05$) was weaker than its effect on perceived ease of use ($p < 0.001$). This can be attributed to the fact that system quality primarily influences users' perception of operational fluency and convenience (i.e., perceived ease of use), whereas judgments of "usefulness" depend on additional factors, such as users' task goals and external requirements, which attenuate the direct effect.

Overall, system quality serves as a prerequisite for user experience but is insufficient to directly drive behavioral intention; its impact is mainly realized indirectly through perceived usefulness and perceived ease of use.

6.2. Information quality

As expected, information quality is a key determinant of both perceived usefulness and perceived ease of use, and it also exerts a direct positive influence on behavioral intention. This finding is consistent with previous studies (Utari & Fauziah, 2023; Yang et al., 2024). In map-based applications, information quality is primarily reflected in dimensions such as spatial and temporal accuracy, correctness, completeness, and consistency (Batini & Scannapieco, 2016). These aspects indicate whether the map can provide users with precise, timely, and logically coherent information.

High information quality can substantially enhance users' perception of system usefulness. This effect has been confirmed in studies on augmented reality and mobile map services (Park & Ohm, 2014;

Yoo, 2020). For AR maps, the core value lies in delivering accurate, up-to-date, and contextually relevant spatial information, such as real-time labeling of attraction locations, dynamic route updates, and supplementary background explanations. When information is accurate, timely, and well-structured, users can directly perceive the practical value of the tool in resolving navigation challenges and enriching the tour experience, thereby judging it as “useful.” Conversely, erroneous, delayed, or redundant information diminishes its perceived utility.

In terms of perceived ease of use, high-quality information presentation (such as concise symbols, clear hierarchies, and context-appropriate display logic) can reduce users’ cognitive load when processing information. By overlaying routes and annotations directly onto the real environment, augmented reality technology reduces the mental effort required for information decoding and matching (Fan et al., 2020), thereby enhancing perceptions of ease of use. In contrast, cluttered annotations and ambiguous guidance increase interpretive difficulty and effectively raise the barrier to use. Prior studies have shown that AR applications employing 3D models and animations can facilitate spatial cognition and mental representation, significantly lowering the cognitive load associated with task execution (Jeffri & Rambli, 2021). This advantage enables users to more easily master navigation operations, thereby further enhancing perceived ease of use.

From the perspective of behavioral intention, precise and high-quality information itself encourages users to adopt AR maps, and this direct effect has been well established in information-centered systems (C.-T. Liu et al., 2023; Nookhao & Kiattisin, 2023; Alsqour et al., 2025; Li & Wang, 2021). High-quality information can meet task requirements, reduce decision-making risk, and enhance users’ sense of psychological security, thereby directly increasing usage intention (Khong et al., 2023). In strongly goal-oriented contexts such as map-based navigation, information itself plays a decisive role in enabling effective decision-making, without necessarily operating through perceived usefulness or perceived ease of use as mediating factors.

6.3. Immersion

This study finds that the effect of immersion on perceived usefulness is not significant; therefore, Hypothesis H7 is not supported. This result is consistent with the findings of Wyssenbach et al. (2025) in the context of VR training. Although immersion is often regarded as a prominent feature of AR and VR technologies, in highly utilitarian contexts such as navigation it does not necessarily translate directly into users’ evaluations of usefulness. This is because immersion is primarily associated with hedonic value, emphasizing users’ emotional experiences and sensory enjoyment during interaction (Nhan et al., 2022), whereas perceived usefulness pertains to utilitarian value and focuses on the extent to which a technology contributes to task performance efficiency (Davis et al., 1989). To be more specific, first, the utilitarian nature of navigation tasks dominates user judgment. In scenarios such as AR navigation where efficiency is paramount, the user’s primary objective is to mitigate spatial uncertainty through precise guidance. Excessive visual effects, while enhancing immersion, may inadvertently distract users and diminish their efficiency in processing critical navigational data. This is consistent with the findings of Tom Dieck et al. (2025), who argue that the acceptance of utilitarian systems depends primarily on whether virtual content facilitates real-world decision-making; the emotional resonance elicited by immersion often misaligns with such functional requirements. Similarly, Makransky and Mayer (2022) observed in educational contexts that while immersion heightens experiential engagement, the core utility of a product remains contingent upon its capacity for precise information transfer. Second, the “attention preemption” effect of the physical environment is particularly pronounced. In settings such as scenic spots, the intrinsic appeal of natural landscapes and cultural heritage often commands the user’s primary attention. Since users are naturally inclined to explore the physical surroundings, the establishment of effective virtual immersion becomes challenging; consequently, it fails to significantly bolster the perception of usefulness (Panou et al., 2018). These factors collectively suggest that the role of immersion within the TAM is heavily dependent on the task environment. In utilitarian contexts where functionality and efficiency are prioritized, immersion serves primarily as a supplementary sensory experience rather than a core driver of perceived usefulness. Consequently, enhancing immersion alone does not significantly influence users’ perceptions of usefulness.

Furthermore, immersion exerts a direct and significant influence on behavioral intention, yet this path is not mediated by PE or PU. This finding aligns with the consumer research conducted by Leveau and Camus (2023). The underlying rationale is that within the context of scenic areas, the functional attributes of AR maps are distinct from their psychological experiential attributes. For users, an AR map serves a dual purpose: it is not only an efficiency-oriented tool for wayfinding but also an experiential medium that facilitates a blended reality environment. When tourists view navigation information naturally overlaid onto real roads and attractions through a mobile camera, they experience immediate pleasure, curiosity, and a desire to explore. These emotional responses represent an instant psychological reward rather than a rational evaluation of instrumental value. Prior studies show that immersion enhances flow experience, which significantly increases user engagement and affective experience, and subsequently translates into higher satisfaction and continued use intention (Ma & He, 2025). In this process, users develop a profound emotional connection with the scene, directly translating virtual interactions into behavioral intentions for sustained usage or recommendation (Di Dalmazi et al., 2024). Users may be driven to continue using AR maps simply because they find the experience “engaging and worth exploring,” irrespective of the system’s operational ease or walking efficiency. Therefore, in scenic contexts, immersion alone can constitute a primary driver of behavioral intention, acting as a direct driver independent of utilitarian judgments.

The results show a positive effect of immersion on PE, consistent with the hypothesis and prior studies (Mishra et al., 2021; Scholz & Smith, 2016). Immersion helps users complete tasks through natural interactions. It reduces the need to switch attention or remember complex steps, resulting in a smoother experience and a greater sense of ease of use. In AR map systems, this effect is particularly pronounced. When route guidance is intuitively overlaid onto real-world surfaces, users can understand navigation intentions without relying on extensive text or complex cognitive conversions, thereby lowering learning costs and cognitive load. Consequently, immersion effectively enhances users’ perceived ease of use of the system.

6.4. Perceived enjoyment

The results indicate that perceived enjoyment has a significant positive effect on perceived usefulness, while its relationship with perceived ease of use is not significant. Additionally, the direct effect of perceived ease of use on behavioral intention was not supported.

Perceived enjoyment refers to the degree of pleasure experienced by users while interacting with a technology (Marzuki et al., 2016). Prior studies have shown that enjoyment can positively influence perceived usefulness (Diefenbach et al., 2014; Li et al., 2024; Sagnier et al., 2020). Specifically, when the AR map incorporates playful elements such as interactive virtual characters, animated stories of attractions, or gamified route displays, users experience enjoyment while using the system. This positive experience encourages users to explore the features more actively. During this process, they recognize the system’s value in navigation and information retrieval, forming the perception that it is both enjoyable and useful. However, the strength of this effect is relatively modest ($p < 0.05$).

In this study, hypothesis H11 was not supported. Although this result contrasts with previous findings (Huang, 2019; Kim et al., 2025), it does not imply that perceived enjoyment has no effect on behavioral intention. Map-based applications are highly utilitarian, and users prioritize efficiently obtaining spatial information and completing navigation tasks. This focus diminishes the direct influence of perceived enjoyment on behavioral intention, which instead operates indirectly through its positive effect on perceived usefulness rather than directly driving usage.

Moreover, perceived enjoyment does not significantly affect perceived ease of use. This finding differs from some prior studies. According to some research, a pleasurable system experience can enhance users’ perception of ease of use (Venkatesh, 2000). Sun and Zhang (2006) also found that in AR contexts, perceived enjoyment had a significant positive effect on perceived ease of use. This discrepancy may be due to AR maps being an emerging technology. Even if the system offers high enjoyment, few users may encounter unfamiliar interfaces or device limitations. These factors offset the positive effect of enjoyment on ease of use, resulting in a non-significant influence of perceived enjoyment on perceived ease of use.

6.5. Perceived usefulness and perceived ease of use

The results show that perceived ease of use has a significant positive effect on perceived usefulness, which is consistent with the conclusions of the TAM model (Davis, 1989). When users find the system easy to operate and the interface user-friendly, they are more likely to form a positive attitude toward its use. In the context of AR maps, intuitive interfaces and smooth interactions reduce the burden of operation and allow users to focus on the travel experience itself. This enhances trust and satisfaction with the system. Such low-barrier and intuitive design not only reduces psychological resistance to adopting new technology but also increases the willingness to continue using the system.

Perceived ease of use and perceived usefulness both have significant positive effects on behavioral intention. In related research, Chen and Tsai (2019) found that in personalized location-based mobile tourism applications (PLMTA) for travel planning, both constructs significantly influenced users' willingness to use the system. When users believe that an AR map system can provide practical support in achieving travel goals, accessing exhibition information, or optimizing route planning, they are more likely to view it as a valuable tool. In tourism contexts, AR maps integrate entertainment with functionality. They meet users' interest while improving travel efficiency, which makes perceived usefulness one of the key factors driving behavioral intention.

6.6. Demographic differences

Demographic analysis reveals that gender, experience, and age moderate several paths in the AR map model.

Specifically, the effects on the $IQ \rightarrow PE$, $IM \rightarrow PE$, and $PE \rightarrow PU$ paths are significantly stronger for female users than for males. Overall, these results suggest that women rely more on experiential cues and process-based judgments. They use these factors to construct their perception of system value during technology adoption. Existing research shows that women are more sensitive to information clarity, structure, and presentation quality (Isenman & Sinclair, 2025). In virtual communities, women specifically hold higher expectations for the quality of information presentation (Liu et al., 2017). Furthermore, they demonstrate high sensitivity toward cognitive effort and understanding costs (M. Ren et al., 2022). Meanwhile, women generally exhibit higher levels of spatial presence and immersion within highly interactive AR contexts (Melo et al., 2022; Putrawangsa et al., 2023). This experience effectively anchors an user's attention to the immediate context, thereby mitigating anxiety by reducing the cognitive load associated with interface complexity and operational steps. Consequently, these immersive qualities are more easily translated into an intuitive judgment that the system is simple to operate. Furthermore, under immersive and interactive conditions, women tend to be more strongly driven by "playfulness" and perceived enjoyment during technology adoption (Chang et al., 2018; Lin & Yeh, 2019). When immersive interactions elicit high levels of emotional resonance and sensory fluency, female users often rely on these positive affective experiences as a basis for evaluating the system's ease of use. This reliance on PE remains significant during the transition from perceived ease of use to perceived usefulness. This finding aligns with established literature suggesting that women prioritize a system's simplicity and operational accessibility more heavily during the adoption process (L. Ren et al., 2022; Zhang et al., 2022). Female users often exhibit higher levels of computer anxiety and lower technological self-efficacy (Ghobadi et al., 2022; Hsieh et al., 2020), a trend that persists in AR navigation contexts involving complex spatial overlays. In these scenarios, a system perceived as easy to operate can effectively mitigate feelings of uncertainty and bolster self-efficacy. This psychological relief, in turn, leads to a more pronounced increase in their evaluation of the system's usefulness (Sobieraj & Krämer, 2020). In contrast, male users tend to assess system utility based on functional outcomes and performance-oriented goals. They demonstrate lower sensitivity to variations in information quality, immersion, and ease of use. This difference explains why the observed moderating effects are generally weaker within the male cohort.

Age significantly moderates the relationship between $PE \rightarrow BI$, with younger users placing a higher premium on ease of use when engaging with AR navigation. This is because being "digital natives" with high technical competence does not translate into a preference for complex systems

(Windasari et al., 2022). On the contrary, they prioritize simple, labor-saving operational experiences to keep pace with fast-moving environments (Pang et al., 2023). Such efficiency allows them to complete tasks rapidly and avoid missing critical information or events (Y. Zou et al., 2023). Conversely, cumbersome operations tend to diminish their interest and engagement with the technology (Perez Vallejos et al., 2021). Consequently, for younger users, user-friendly interfaces and streamlined procedures effectively lower the barrier to adoption, thereby strengthening behavioral intention (Chang et al., 2015; Penney et al., 2021). Furthermore, this demographic emphasizes the integration of social interaction with entertainment during technology use. In this context, a “convenient and effortless” experience becomes a critical prerequisite for sustained usage and peer recommendation (Kusairi et al., 2025). In contrast, older users demonstrate a pronounced pragmatic orientation when adopting new technologies like AR navigation. They prioritize the functional value and necessity of the technology, focusing on its practical impact on efficiency and quality of life (An et al., 2024; Schroeder et al., 2023; Yap et al., 2022). For this demographic, the experiential aspect of the interaction is secondary to its utility. Consequently, older users are more likely to abandon a system if it fails to provide a clear and compelling justification for its practical use (Hauk et al., 2018).

The moderating effects on the $SQ \rightarrow PU$, $IQ \rightarrow BI$, $PU \rightarrow BI$, and $PE \rightarrow PU$ paths are significantly stronger for inexperienced users than for experienced ones. This disparity arises because, in the absence of prior knowledge, learning a new system demands substantial cognitive resources. Consequently, inexperienced users are more susceptible to system quality factors, such as interface clarity, navigational efficiency, and operational convenience (Shao et al., 2020). Accordingly, they demonstrate a stronger preference for technologies that are perceived as easy to operate (Venkatesh & Davis, 2000). Systems that are difficult to operate increase learning time and diminish self-efficacy. This elevation in cognitive load and frustration prevents users from focusing on the system’s functional value (Baki et al., 2021), thereby reducing their perception of its usefulness. Consequently, the practical utility of the technology only becomes apparent to first-time users when the system provides clear feedback and fluid interaction (Dutta et al., 2022; Su et al., 2023). Therefore, during the initial usage stage, SQ and PE exert a more significant influence on PU. As users gain experience and deepen their understanding of the system, their perception of usefulness becomes increasingly dependent on their own technical proficiency (Han & Guo, 2025). At this stage, PU is primarily determined by whether the system’s functionalities can deliver tangible performance gains (Karunarathne & Abeyratne, 2020). In other words, for experienced users, the evaluation of PU shifts toward the instrumental value of the technology, rather than being mediated by its ease of use. The moderating effect on BI is most pronounced among inexperienced users, suggesting that novices rely heavily on perceived usefulness and immediate informational cues. According to the Expectation-Confirmation Model (ECM), perceived usefulness is the primary belief driving intent during early adoption (Bhattacharjee, 2001). For AR navigation novices who lack the domain expertise or benchmarks to evaluate “utility” objectively, a system that demonstrates effective spatial guidance can more readily trigger strong usage intentions. Poor information quality in AR annotations can trigger cognitive overload in novices, leading to early abandonment (Mudau et al., 2024). As usage experience increases, users gradually take a broader range of factors into account, rather than focusing solely on task completion capability. Moreover, because experienced users possess richer prior knowledge and well-developed cognitive schemas, they rely less on fine-grained informational cues during AR navigation. They show greater tolerance for localized information deficiencies and are more likely to compensate through alternative strategies rather than abandoning use altogether (Laumer et al., 2017). Therefore, the moderating effects on the $PU \rightarrow BI$ and $IQ \rightarrow BI$ paths are significantly amplified among inexperienced users.

7. Conclusion, limitations and future directions

This study investigated the factors influencing tourists’ use of AR-based real-world maps based on an extended TAM framework. The findings provide practical guidance for the design and optimization of AR maps. Based on the results, the development of AR real-world maps should consider multiple dimensions, including system quality, information quality, immersion, perceived enjoyment, perceived

ease of use, perceived usefulness, and demographic characteristics, in order to enhance user experience and intention to use.

First, at the level of system and information quality, priority should be given to ensuring system stability, fast response, and an intuitive interface. Layout and positioning accuracy should be optimized to reduce learning costs, allowing users to quickly become proficient and enhancing their trust in the product. Information content should be accurate, complete, and hierarchically clear, with basic navigation information presented first. The use of AR virtual annotations can improve visual clarity and support users' navigation and knowledge acquisition needs. Second, regarding immersion and perceived enjoyment, immersive experience can be enhanced by optimizing the integration of virtual and real elements, implementing gesture and voice interactions, and providing dynamic route guidance and contextual content. These features improve operational fluency and reduce cognitive load, thereby increasing perceived ease of use. Enjoyment can be enhanced by incorporating gamification elements, virtual characters, dynamic effects, or social features, which in turn improves perceived usefulness and indirectly strengthens users' intention to use. In the design of ease of use and usefulness, simple and intuitive symbols, one-click operations, and other low-barrier designs should be adopted. High-value functions, such as real-time congestion avoidance, personalized routing, and in-depth explanations of attractions, should be provided so that ease of use and usefulness mutually reinforce each other, ultimately enhancing behavioral intention. In addition, demographic differences should be taken into account. For female users, developers should enhance sensory fluency through structured information and immersive interactions to mitigate anxiety and bolster perceived ease of use. Conversely, for male users, the focus should remain on functional output and the reduction of non-essential interactions. For older demographics, the design must prioritize utilitarian functions with simplified workflows, using large fonts and high-contrast interfaces to align with their pragmatic orientation. For younger users, optimizing interface friendliness and task efficiency is essential to satisfy their preference for convenient experiences. Finally, for novices, guided onboarding and "beginner modes" are necessary to lower cognitive load. In contrast, experienced users should be provided with advanced features and tiered functionality to support their demand for performance enhancement.

It should be noted that this study has several limitations. First, the data were collected from visitors to the Detian Transnational Waterfall Scenic Area in China, which may limit the external validity of the findings. Future research may collect data across different countries, regions, or types of attractions to verify the generalizability of the results. Such studies could also explore the influence of cultural or environmental factors on behavioral intention toward AR maps. Second, this study relied on self-reported data through questionnaires, which may be subject to cognitive bias or social desirability effects, leading to discrepancies between perceived and actual usage behavior. Future studies may combine experimental or field-based behavioral data, such as AR map usage logs, eye-tracking measures, or physiological responses, to obtain more objective assessments of user experience.

Acknowledgment

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Author contributions

CRedit: **Shuo Li**: Conceptualization, Data curation, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing; **Yuhui Wang**: Conceptualization, Project administration, Writing – review & editing; **Lei Liu**: Project administration, Supervision, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Shuo Li  <http://orcid.org/0009-0001-0548-9015>

Yuhui Wang  <http://orcid.org/0000-0003-3748-3723>

Lei Liu  <http://orcid.org/0009-0003-8399-0168>

References

- Abdel Rady, H., & Khalf, A. (2019). Towards smart tourism destination: An empirical study on Sharm El Sheikh City, Egypt. *International Journal of Heritage, Tourism and Hospitality*, 13(1), 78–95. <https://doi.org/10.21608/ijhth.2019.35106>
- Aggag, A., & Kortam, W. (2025). Using augmented reality to improve tourism marketing effectiveness. *Sustainability*, 17(13), 5747. <https://doi.org/10.3390/su17135747>
- Alarifi, A., Al-Salman, A., Alsaleh, M., Alnafessah, A., Al-Hadhrani, S., Al-Ammar, M. A., & Al-Khalifa, H. S. (2016). Ultra wideband indoor positioning technologies: Analysis and recent advances. *Sensors*, 16(5), 707. <https://doi.org/10.3390/s16050707>
- Alsqour, M., Attar, H., Alibraheem, M. H., Alsaqoor, S., & Alsaleem, E. (2025). System quality, information quality, perceived usefulness, and ease of use as determinants of intention to continue using a digital accounting system among Jordanian SMEs. *Engineering Management in Production and Services*, 17(2), 78–89. <https://doi.org/10.2478/emj-2025-0013>
- Álvarez-Marín, A., Velázquez-Iturbide, J. Á., Castillo-Vergara, M., & Acuña-Opazo, C. (2025). The moderating role of aesthetics and information quality for acceptance of augmented reality. *Innovations in Education and Teaching International*, 62(1), 102–119. <https://doi.org/10.1080/14703297.2023.2267509>
- Amirian, P., Basiri, A., Gales, G., Winstanley, A., & McDonald, J. (2015). The next generation of navigational services using OpenStreetMap data: The integration of augmented reality and graph databases. In J. Jokar Arsanjani, A. Zipf, P. Mooney, & M. Helbich (Eds.), *OpenStreetmap in GIScience: Experiences, research, and applications* (pp. 211–228). Springer. https://doi.org/10.1007/978-3-319-14280-7_11
- An, J., Zhu, X., Wan, K., Xiang, Z., Shi, Z., An, J., & Huang, W. (2024). Older adults' self-perception, technology anxiety, and intention to use digital public services. *BMC Public Health*, 24(1), 3533. <https://doi.org/10.1186/s12889-024-21088-2>
- Asraf, S. M. H., Hashim, A. F. M., & Idrus, S. Z. S. (2020, April). *Mobile application outdoor navigation using location-based augmented reality (AR)* [Paper presentation]. Journal of Physics: Conference Series (vol. 1529, no. 2, p. 022098). IOP Publishing. <https://doi.org/10.1088/1742-6596/1529/2/022098/meta>
- Azuma, R. T. (1997). A survey of augmented reality. *Presence: Teleoperators and Virtual Environments*, 6(4), 355–385. <https://doi.org/10.1162/pres.1997.6.4.355>
- Baki, R., Birgoren, B., & Aktepe, A. (2021). Identifying factors affecting intention to use in distance learning systems. *Turkish Online Journal of Distance Education*, 22(2), 58–80. <https://doi.org/10.17718/TOJDE.906545>
- Bashir, M., Asghar, S., Ashfaq, M., Raza, S. S., & Zahoor, S. (2024). Digitalizing tourism and interactive navigation: A case study of Pakistan. *Journal of Asian Development Studies*, 13(1), 547–566. <https://doi.org/10.62345/jads.2024.13.1.46>
- Batini, C., & Scannapieco, M. (2016). Information quality dimensions for maps and texts. In C. Batini, & M. Scannapieco (Eds.), *Data and information quality: Dimensions, principles and techniques* (pp. 53–86). Springer. https://doi.org/10.1007/978-3-319-24106-7_3
- Bhattacharjee, A. (2001). Understanding information systems continuance: An expectation-confirmation model. *MIS Quarterly*, 25(3), 351–370. <https://doi.org/10.2307/3250921>
- Brata, K. C., Funabiki, N., Panduman, Y. Y. F., & Fajrianti, E. D. (2024). An enhancement of outdoor location-based augmented reality anchor precision through VSLAM and Google Street View. *Sensors*, 24(4), 1161. <https://doi.org/10.3390/s24041161>
- Briz, M. (2021). The role of leisure industry in municipal economic growth. *International Journal of Academe and Industry Research*, 2(4), 16–31. <https://doi.org/10.53378/352081>
- Byrne, B. M. (2001). Structural equation modeling with AMOS, EQS, and LISREL: Comparative approaches to testing for the factorial validity of a measuring instrument. *International Journal of Testing*, 1(1), 55–86. https://doi.org/10.1207/S15327574IJT0101_4
- Cabeleira, M., & Vaz de Carvalho, C. (2025). Using augmented reality to improve touristic efficacy. *Computers*, 14(2), 75. <https://doi.org/10.3390/computers14020075>
- Calisir, F., Altin Gumussoy, C., Bayraktaroglu, A. E., & Karaali, D. (2014). Predicting the intention to use a web-based learning system: Perceived content quality, anxiety, perceived system quality, image, and the technology acceptance model. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 24(5), 515–531. <https://doi.org/10.1002/hfm.20548>
- Cao, J., Lam, K.-Y., Lee, L.-H., Liu, X., Hui, P., & Su, X. (2023). Mobile augmented reality: User interfaces, frameworks, and intelligence. *ACM Computing Surveys*, 55(9), 1–36. <https://doi.org/10.1145/3557999>

- Cardoso, I. B. F., & Ruiz, T. C. D. (2021). Smart tourism destinations - a case study of Seoul, South Korea. *Applied Tourism*, 6(1), 36–44. <https://doi.org/10.14210/at.v6n1>
- Carmigniani, J., Furht, B., Anisetti, M., Ceravolo, P., Damiani, E., & Ivkovic, M. (2011). Augmented reality technologies, systems and applications. *Multimedia Tools and Applications*, 51(1), 341–377. <https://doi.org/10.1007/s11042-010-0660-6>
- Chai-Arayalert, S., Sansompron, N., Chodchuang, S., Chantanaphant, J., & Duangjan, C. (2023). Digital platform-mediated tourism system in small-town destination. *International Journal of Interactive Mobile Technologies*, 17(05), 143–161. <https://doi.org/10.3991/ijim.v17i05.36835>
- Chang, C.-W., Heo, J., Yeh, S.-C., Han, H.-Y., & Li, M. (2018). The effects of immersion and interactivity on college students' acceptance of a novel VR-supported educational technology for mental rotation. *IEEE Access*, 6(1), 66590–66599. <https://doi.org/10.1109/ACCESS.2018.2878270>
- Chang, S.-C., Sun, C.-C., Pan, L.-Y., & Wang, M.-Y. (2015). An extended TAM to explore behavioural intention of consumers to use M-commerce. *Journal of Information & Knowledge Management*, 14(02), 1550014. <https://doi.org/10.1142/S0219649215500148>
- Cheliotis, K., Liarokapis, F., Kokla, M., Tomai, E., Pastra, K., Anastopoulou, N., Bezerianou, M., Darra, A., & Kavouras, M. (2023). A systematic review of application development in augmented reality navigation research. *Cartography and Geographic Information Science*, 50(3), 249–271. <https://doi.org/10.1080/15230406.2023.2194032>
- Chen, C.-C., & Tsai, J.-L. (2019). Determinants of behavioral intention to use the personalized location-based mobile tourism application: An empirical study by integrating TAM with ISSM. *Future Generation Computer Systems*, 96(1), 628–638. <https://doi.org/10.1016/j.future.2017.02.028>
- Chen, J., Wu, X., & Lai, I. K. W. (2023). A systematic literature review of virtual technology in hospitality and tourism (2013–2022). *Sage Open*, 13(3), 1–18. <https://doi.org/10.1177/21582440231193297>
- Cheng, A., Ma, D., Pan, Y., & Qian, H. (2024). Enhancing museum visiting experience: Investigating the relationships between augmented reality quality, immersion, and TAM using PLS-SEM. *International Journal of Human-Computer Interaction*, 40(17), 4521–4532. <https://doi.org/10.1080/10447318.2023.2227832>
- Cheverst, K., Davies, N., Mitchell, K., Friday, A., & Efstratiou, C. (2000). Developing a context-aware electronic tourist guide: Some issues and experiences. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '00)* (pp. 17–24). ACM. <https://doi.org/10.1145/332040.332047>
- Çilkin, R. E., & Toksöz, D. (2024). Reflections of technological developments on tourist guidance: Mobile tourist guide applications. *Journal of Tourism & Gastronomy Studies*, 12(1), 445–455. <https://doi.org/10.21325/jotags.2024.1388>
- Cynthia, M. T., Ingrid, P. L., & Alicia, Y. M. (2021). Digitization trends in hospitality and tourism. *Smart Tourism*, 2(2), 1–11. <https://doi.org/10.54517/st.v2i2.1709>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>
- Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1989). User acceptance of computer technology: A comparison of two theoretical models. *Management Science*, 35(8), 982–1003. <https://doi.org/10.1287/mnsc.35.8.982>
- de Oliveira, I. J. (2005). A cartografia aplicada ao planejamento do turismo. *Boletim Goiano de Geografia*, 25(1), 29–46. <https://doi.org/10.5216/bgg.v25i1.4109>
- DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems*, 19(4), 9–30. <https://doi.org/10.1080/07421222.2003.11045748>
- Di Dalmazi, M., Mandolfo, M., Guixeres, J., Alcañiz Raya, M., & Lamberti, L. (2024). How immersive technologies impact behavioral responses in destination marketing: The role of physiological arousal, presence, and age. *International Journal of Contemporary Hospitality Management*, 36(11), 3628–3650. <https://doi.org/10.1108/IJCHM-09-2023-1511>
- Dickmann, F. (2005). Effectiveness and efficiency of tourism maps in the world wide web and their potential for mobile map services. In L. Meng, T. Reichenbacher, & A. Zipf (Eds.), *Map-based mobile services: Theories, methods and implementations* (pp. 43–53). Springer. https://doi.org/10.1007/3-540-26982-7_4
- Diefenbach, S., Kolb, N., & Hassenzahl, M. (2014). The 'hedonic' in human-computer interaction: History, contributions, and future research directions [Paper presentation]. Proceedings of the 2014 Conference on Designing Interactive Systems (pp. 305–314). ACM. <https://doi.org/10.1145/2598510.2598549>
- Dutta, R., Mantri, A., & Singh, G. (2022). Evaluating system usability of mobile augmented reality application for teaching Karnaugh-Maps. *Smart Learning Environments*, 9(1), 6. <https://doi.org/10.1186/s40561-022-00189-8>
- Eboy, O. V. (2017). Tourism mapping: An overview of cartography and the use of GIS. *BIMP-EAGA Journal for Sustainable Tourism Development*, 6(1), 61–67. <https://doi.org/10.51200/bimpeagajtsd.v6i1.3068>
- Evangelou, A., Klefodimos, A., & Lappas, G. (2024). Creating location-based mobile applications for tourism: A virtual AR guide for Western Macedonia. *Digital*, 4(1), 271–301. <https://doi.org/10.3390/digital4010014>
- Fan, X., Chai, Z., Deng, N., & Dong, X. (2020). Adoption of augmented reality in online retailing and consumers' product attitude: A cognitive perspective. *Journal of Retailing and Consumer Services*, 53(1), 101986. <https://doi.org/10.1016/j.jretconser.2019.101986>

- Feiner, S., MacIntyre, B., Höllerer, T., & Webster, A. (1997). A touring machine: Prototyping 3D mobile augmented reality systems for exploring the urban environment. *Personal Technologies*, 1(4), 208–217. <https://doi.org/10.1007/BF01682023>
- Fitria, F., Yahya, M., Ali, M. I., Purnamawati, P., & Mappalotteng, A. M. (2024). The impact of system quality and user satisfaction: The mediating role of ease of use and usefulness in e-learning systems. *International Journal of Environment, Engineering and Education*, 6(2), 119–131. <https://doi.org/10.55151/ijeedu.v6i2.134>
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50. <https://doi.org/10.1177/002224378101800104>
- Gajdošík, T. (2018). Smart tourism: Concepts and insights from Central Europe. *Czech Journal of Tourism*, 7(1), 25–44. <https://doi.org/10.1515/cjot-2018-0002>
- Gao, Y., She, M., & Zhang, X. (2023). Research on application of multi-navigation technology fusion to improve vehicle positioning accuracy in complex scenes [Paper presentation]. 2023 IEEE 5th International Conference on Civil Aviation Safety and Information Technology (ICCASIT) (pp. 263–268). IEEE. <https://doi.org/10.1109/ICCASIT58768.2023.10351502>
- Geiger, P., Schickler, M., Pryss, R., Schobel, J., & Reichert, M. (2014). Location-based mobile augmented reality applications – challenges, examples, lessons learned [Paper presentation]. Proceedings of the 10th International Conference on Web Information Systems and Technologies (pp. 383–394). SciTePress. <https://doi.org/10.5220/0004975503830394>
- Gerstweiler, G., Vonach, E., & Kaufmann, H. (2015). HyMoTrack: A mobile AR navigation system for complex indoor environments. *Sensors*, 16(1), 17. <https://doi.org/10.3390/s16010017>
- Ghandour, A., Kintonova, A., Demidchik, N., & Sverdlkova, E. (2021). Solving tourism management challenges by means of mobile augmented reality applications. *International Journal of Web-Based Learning and Teaching Technologies*, 16(6), 1–16. <https://doi.org/10.4018/IJWLTT.293280>
- Ghobadi, M., Shirowzhan, S., Ghiai, M. M., Mohammad Ebrahimzadeh, F., & Tahmasebinia, F. (2022). Augmented reality applications in education and examining key factors affecting the users' behaviors. *Education Sciences*, 13(1), 10. <https://doi.org/10.3390/educsci13010010>
- Gozali, A. A., Prawita, F. H. N., Gumilar, I., Putra, H. R. R., & Fauzan, M. A. (2024). Geo-navigation in museums: Augmented reality application in the Geological Museum Indonesia. *Jurnal Infotel*, 16(1), 156–172. <https://doi.org/10.20895/infotel.v16i1.1103>
- Gudadhe, M., Bhanse, D., Dhengre, B., Patle, T., Mange, K., & Patil, V. (2021). Augmented reality in tourism: An innovative way to attract tourists. *International Journal of Innovations in Engineering and Science*, 6(6), 01–04. <https://doi.org/10.46335/IJIES.2021.6.6.1>
- Guerra, F., & Boutoura, C. (2001). An electronic lens on digital tourist city-maps. In L. Li (Ed.), *Mapping the 21st century: Proceedings of the 20th International Cartographic Conference* (pp. 1151–1157). Chinese Society of Geodesy, Photogrammetry, and Cartography.
- Guo, H., Yang, N., Wang, Z., & Fang, H. (2023). Effects of spatial reference frames, map dimensionality, and navigation modes on spatial orientation efficiency. *ISPRS International Journal of Geo-Information*, 12(12), 476. <https://doi.org/10.3390/ijgi12120476>
- Gupta, A., & Dogra, N. (2017). Tourist adoption of mapping apps: A UTAUT2 perspective of smart travellers. *Tourism and Hospitality Management*, 23(2), 145–161. <https://doi.org/10.20867/thm.23.2.6>
- Gupta, S., & Sharma, A. (2024). Understanding the role of personalisation in augmented reality applications for enhancing user experience. *NMIMS Management Review*, 32(3), 197–210. <https://doi.org/10.1177/09711023241303208>
- Hair, J. F., Tatham, R. L., Anderson, R. E., & Black, W. (1998). Multivariate data analysis. *Technometrics*, 30(1), 130–131. <https://doi.org/10.1080/00401706.1988.10488353>
- Hair, J. F., Sarstedt, M., Ringle, C. M., & Mena, J. A. (2012). An assessment of the use of partial least squares structural equation modeling in marketing research. *Journal of the Academy of Marketing Science*, 40(3), 414–433. <https://doi.org/10.1007/s11747-011-0261-6>
- Han, F., & Guo, J. (2025). How does university students' academic major (STEM vs. non-STEM) affect their acceptance of e-learning: A multi-group analysis. *International Journal of Educational Technology in Higher Education*, 22(1), 41. <https://doi.org/10.1186/s41239-025-00543-z>
- Happ, É., & Ivancsó-Horváth, Z. (2018). Digital tourism is the challenge of future—a new approach to tourism. *Knowledge Horizons-Economics*, 10(2), 9–16. <https://www.orizonturi.ucdc.ro/arhiva/KHE%20nr.%202%20-%202018/1.%20DIGITAL%20TOURISM%20IS%20THE%20CHALLENGE%20OF%20FUTURE.pdf>
- Harede, B. M., & Fekry Ibrahim, W. (2019). An experience of presenting cultural attractions and tourism services using Google maps technology in Matrouh Governorate. *Journal of Association of Arab Universities for Tourism and Hospitality*, 17(2), 69–81. <https://doi.org/10.21608/jaauth.2019.76469>
- Hassenzahl, M. (2018). The thing and I: Understanding the relationship between user and product. In *Funology 2: From usability to enjoyment* (pp. 301–313). Springer. <https://doi.org/10.1007/978-3-319-68213-6>
- Hauk, N., Hüffmeier, J., & Krumm, S. (2018). Ready to be a silver surfer? A meta-analysis on the relationship between chronological age and technology acceptance. *Computers in Human Behavior*, 84(1), 304–319. <https://doi.org/10.1016/j.chb.2018.01.020>

- He, Y., & Wang, W. (2025). Factors influencing visitors' use of augmented reality technology in museum guided tours. *PLOS One*, 20(10), e0332688. <https://doi.org/10.1371/journal.pone.0332688>.
- Hsieh, Y. C., Tsai, W. C., & Hsia, Y. C. (2020, July). *A study on technology anxiety among different ages and genders* [Paper presentation]. International Conference on Human-Computer Interaction, Cham (pp. 241–254). Springer International Publishing. https://doi.org/10.1007/978-3-030-50232-4_17
- Huang, J., Kinatader, M., Dunn, M. J., Jarosz, W., Yang, X.-D., & Cooper, E. A. (2019). An augmented reality sign-reading assistant for users with reduced vision. *PLOS One*, 14(1), e0210630. <https://doi.org/10.1371/journal.pone.0210630>
- Huang, Y.-C., Backman, S. J., Backman, K. F., & Moore, D. (2013). Exploring user acceptance of 3D virtual worlds in travel and tourism marketing. *Tourism Management*, 36(1), 490–501. <https://doi.org/10.1016/j.tourman.2012.09.009>
- Huang, Y.-M. (2019). Exploring students' acceptance of educational computer games from the perspective of learning strategy. *Australasian Journal of Educational Technology*, 35(3), 132–149. <https://doi.org/10.14742/ajet.3330>
- Huynh, B., Ibrahim, A., Chang, Y. S., Höllerer, T., & O'Donovan, J. (2019). User perception of situated product recommendations in augmented reality. *International Journal of Semantic Computing*, 13(03), 289–310. <https://doi.org/10.1142/S1793351X19400129>
- İnci, B. (2017). Consumer attitudes towards private shopping sites in Turkey. *Chinese Business Review*, 16(1), 1–18. <https://doi.org/10.17265/1537-1506/2017.01.001>
- Isenman, L., & Sinclair, M. (2025). Women, intuition, and management—the Yin and Yang of nonconscious thought. *Frontiers in Psychology*, 16(1), 1572888. <https://doi.org/10.3389/fpsyg.2025.1572888>
- Jalilvand, M. R., & Ghasemi, H. (2024). Augmented reality technology in tourism and hospitality research: A review from 2010 to 2024. *Journal of Science and Technology Policy Management*, 15(4), 1–27. <https://doi.org/10.1108/JSTPM-04-2024-0136>
- Jeffri, N. F. S., & Rambli, D. R. A. (2021). A review of augmented reality systems and their effects on mental workload and task performance. *Heliyon*, 7(3), e06277. <https://doi.org/10.1016/j.heliyon.2021.e06277>
- Jiang, S., Moyle, B., Yung, R., Tao, L., & Scott, N. (2023). Augmented reality and the enhancement of memorable tourism experiences at heritage sites. *Current Issues in Tourism*, 26(2), 242–257. <https://doi.org/10.1080/13683500.2022.2026303>
- Jiman, J., & Kulal, S. M. (2023). *Augmented reality (AR) and virtual reality (VR) applications in tourism: Embracing emerging technologies for improved tourist experiences in Malaysian tourism industry* [Paper presentation]. Proceedings of the International Conference on Digital Advanced Tourism, Management, and Technology, Semarang, Indonesia (Vol. 1, no. 2, pp. 188–200). <https://doi.org/10.56910/ictmt.v1i2.34>
- Jo, H., & Bang, Y. (2023). Understanding continuance intention of enterprise resource planning (ERP): TOE, TAM, and IS success model. *Heliyon*, 9(10), e21019. <https://doi.org/10.1016/j.heliyon.2023.e21019>
- Jung, T., Chung, N., & Leue, M. C. (2015). The determinants of recommendations to use augmented reality technologies: The case of a Korean theme park. *Tourism Management*, 49(1), 75–86. <https://doi.org/10.1016/j.tourman.2015.02.013>
- Kabadayı, M. (2020). Otel işletmelerinde sanal gerçeklik ve artırılmış gerçeklik uygulamaları. *Seyahat ve Otel İşletmeciliği Dergisi*, 17(3), 464–479. <https://doi.org/10.24010/soid.696483>
- Karaş, Y. K., Atay, L., Sevinç, H. K., & Duru, A. (2021). Investigation of GIS-based mobile route planning–navigation applications for tourism activities. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVI-4/W5-2021(1), 321–326. <https://doi.org/10.5194/isprs-archives-XLVI-4-W5-2021-321-2021>
- Karunaratne, E., & Abeyratne, A. (2020). Role of employee training and experience on the adaption of computerized maintenance management system. *Strategic Management*, 25(4), 3–16. <https://doi.org/10.5937/StraMan2004003K>
- Kaźmierczak, R., Szczepańska, A., Kowalczyk, C., Grunwald, G., & Janowski, A. (2021). Using AR technology in tourism based on the example of maritime educational trips—a conceptual model. *Sustainability*, 13(13), 7172. <https://doi.org/10.3390/su13137172>
- Khong, I., Yusuf, N. A., Nuriman, A., & Yadila, A. B. (2023). Exploring the impact of data quality on decision-making processes in information intensive organizations. *APTISI Transactions on Management (ATM)*, 7(3), 253–260. <https://doi.org/10.33050/atm.v7i3.2138>
- Kim, M., & Doeller, C. F. (2022). Adaptive cognitive maps for curved surfaces in the 3D world. *Cognition*, 225(1), 105126. <https://doi.org/10.1016/j.cognition.2022.105126>
- Kim, M., & Doeller, C. F. (2024). Cognitive maps for a non-Euclidean environment: Path integration and spatial memory on a sphere. *Psychological Science*, 35(11), 1217–1230. <https://doi.org/10.1177/09567976241279291>
- Kim, Y. W., Cha, M. C., Yoon, S. H., & Lee, S. C. (2025). Not merely useful but also amusing: Impact of perceived usefulness and perceived enjoyment on the adoption of AI-powered coding assistant. *International Journal of Human-Computer Interaction*, 41(10), 6210–6222. <https://doi.org/10.1080/10447318.2024.2375701>
- Kline, R. B. (2023). *Principles and practice of structural equation modeling*. The Guilford Publications.

- Koga, Y., & Yamamoto, K. (2021). A sightseeing planning support system with gamification. *Journal of Geographic Information System*, 13(04), 485–507. <https://doi.org/10.4236/jgis.2021.134027>
- Kostin, K. B. (2018). Foresight of the global digital trends. *Strategic Management*, 23(2), 11–19. <https://doi.org/10.5937/StraMan1801011K>
- Kusairi, K., Sukmawati, A., A.s, N., & Rahman, M. S. (2025). Predicting M-banking adoption: The moderating role of age in technology acceptance. *Cogent Business & Management*, 12(1), 2547964. <https://doi.org/10.1080/23311975.2025.2547964>
- Lai, Z. J., Leong, M. K., Khoo, K. L., & Sidhu, S. K. (2025). Integrating technology acceptance model and value-based adoption model to determine consumers' perception of value and intention to adopt AR in online shopping. *Asia Pacific Journal of Marketing and Logistics*, 37(1), 1–19. <https://doi.org/10.1108/APJML-03-2024-0386>
- Laumer, S., Maier, C., & Weitzel, T. (2017). Information quality, user satisfaction, and the manifestation of work-arounds: A qualitative and quantitative study of enterprise content management system users. *European Journal of Information Systems*, 26(4), 333–360. <https://doi.org/10.1057/s41303-016-0029-7>
- Laurini, R., Gordillo, S., Raffort, F., Servigne, S., Rossi, G., Wang, N., & Fortier, A. (2008). A primer of picture-aided navigation in mobile systems. In A. Ruas & C. Gold (Eds.), *Headway in spatial data handling* (pp. 595–611). Springer. https://doi.org/10.1007/978-3-540-68566-1_34
- Lee, H.-H., & Sung, H.-C. (2023). Unveiling the confirmation factors of information system quality on continuance intention towards online cryptocurrency exchanges: The extension of the expectation confirmation model. *Information*, 14(9), 482. <https://doi.org/10.3390/info14090482>
- Leveau, P. H., & Camus, E. S. (2023). Embodiment, immersion, and enjoyment in virtual reality marketing experiences. *Psychology & Marketing*, 40(7), 1329–1343. <https://doi.org/10.1002/mar.21822>
- Li, J., Tu, X.-F., Nie, J.-W., Ye, J., & Lv, C. (2024). Visitors' acceptance of wearable AR technology in museums. *Sage Open*, 14(4), 21582440241303507. <https://doi.org/10.1177/21582440241303507>
- Li, Y., & Wang, J. (2021). Evaluating the impact of information system quality on continuance intention toward cloud financial information system. *Frontiers in Psychology*, 12(1), 713353. <https://doi.org/10.3389/fpsyg.2021.713353>
- Lin, P.-H., & Yeh, S.-C. (2019). How motion-control influences a VR-supported technology for mental rotation learning: From the perspectives of playfulness, gender difference and technology acceptance model. *International Journal of Human-Computer Interaction*, 35(18), 1736–1746. <https://doi.org/10.1080/10447318.2019.1571784>
- Liu, C.-T., Guo, Y. M., & Hsu, J.-L. (2023). Creating and validating an information quality scale for e-commerce platforms. *Journal of Electronic Commerce in Organizations*, 21(1), 1–28. <https://doi.org/10.4018/JECO.327350>
- Liu, Y., Li, Y., Zhang, H., & Huang, W. W. (2017). Gender differences in information quality of virtual communities: A study from an expectation-perception perspective. *Personality and Individual Differences*, 104(1), 224–229. <https://doi.org/10.1016/j.paid.2016.08.011>
- Liu, Y., Sun, J. C.-Y., & Chen, S.-K. (2023). Comparing technology acceptance of AR-based and 3D map-based mobile library applications: A multigroup SEM analysis. *Interactive Learning Environments*, 31(7), 4156–4170. <https://doi.org/10.1080/10494820.2021.1955271>
- Lu, F., Zhou, H., Guo, L., Chen, J., & Pei, L. (2021). An ARCore-based augmented reality campus navigation system. *Applied Sciences*, 11(16), 7515. <https://doi.org/10.3390/app11167515>
- Lu, M., Arikawa, M., Oba, K., Ishikawa, K., Jin, Y., Utsumi, T., & Sato, R. (2024). Indoor AR navigation framework based on geofencing and image-tracking with accumulated error correction. *Applied Sciences*, 14(10), 4262. <https://doi.org/10.3390/app14104262>
- Ma, W., Zhang, S., & Huang, J. (2021). Mobile augmented reality based indoor map for improving geo-visualization. *PeerJ. Computer Science*, 7(1), e704. <https://doi.org/10.7717/peerj-cs.704>
- Ma, Y., & He, W. (2025). “Coolness” and “joy” in games: Factors influencing mobile game players' willingness to make in-game purchases. *Asia Pacific Journal of Marketing and Logistics*, 37(2), 331–348. <https://doi.org/10.1108/APJML-04-2024-0539>
- Makransky, G., & Mayer, R. E. (2022). Benefits of taking a virtual field trip in immersive virtual reality: Evidence for the immersion principle in multimedia learning. *Educational Psychology Review*, 34(3), 1771–1798. <https://doi.org/10.1007/s10648-022-09675-4>
- Maria, A. D., Octafian, R., Pranoto, P., & Mistriani, N. (2024). Augmented reality for smart tourism: A systematic review. *Proceeding of The International Global Tourism Science and Vocational Education*, 1(1), 01–09. <https://doi.org/10.62951/icgtsave.v1i1.1>
- Marian-Vladut, T., Octavian, T. C., & Paul, P. (2025). Evaluating user acceptance and usability of AR-based indoor navigation in a university setting: An empirical study. *International Journal of Advanced Computer Science and Applications*, 16(4), 49–56. <https://doi.org/10.14569/IJACSA.2025.0160406>
- Marzuki, M. I. I., Rosly, A. N., Roslan, N. S., Abdullah, D., Kamal, S. B. M., & Azmi, A. (2016). The role of perceived interactivity, perceived ease of use, perceived usefulness, and perceived enjoyment toward intention to use online mapping service applications. *International Academic Research Journal of Business and Technology*, 2(2), 135–139. https://www.iarjournal.com/wp-content/uploads/IARJBT2016_2_135-139.pdf

- Melo, M., Coelho, H., Gonçalves, G., Losada, N., Jorge, F., Teixeira, M. S., & Bessa, M. (2022). Immersive multi-sensory virtual reality technologies for virtual tourism: A study of the user's sense of presence, satisfaction, emotions, and attitudes. *Multimedia Systems*, 28(3), 1027–1037. <https://doi.org/10.1007/s00530-022-00898-7>
- Metz, D. (2022). The impact of digital navigation on travel behaviour. *UCL Open. Environment*, 4(1), e034. <https://doi.org/10.14324/111.444/ucloe.000034>
- Mieli, M., Zillinger, M., & Nilsson, J.-H. (2024). Phygital time geography, or: What about technology in tourists' space-time behaviour? *Tourism Geographies*, 26(3), 389–406. <https://doi.org/10.1080/14616688.2024.2318722>
- Milgram, P., Takemura, H., Utsumi, A., & Kishino, F. (1995). Augmented reality: A class of displays on the reality-virtuality continuum. In J. O. Merriitt & M. T. Bolas (Eds.), *Stereoscopic displays and virtual reality systems II* (Vol. 2351, pp. 282–292). SPIE. <https://doi.org/10.1117/12.197321>
- Mishra, A., Shukla, A., Rana, N. P., & Dwivedi, Y. K. (2021). From “touch” to a “multisensory” experience: The impact of technology interface and product type on consumer responses. *Psychology & Marketing*, 38(3), 385–396. <https://doi.org/10.1002/mar.21436>
- Moss, J. M. (2015). *Maps, tourism, and historical pedagogy: A study of power, identity, and the politics of representation in two southern cities* [Doctoral dissertation]. ScholarWorks@ Georgia State University.
- Mudau, T. N., Cohen, J., & Papageorgiou, E. (2024). Determinants and consequences of routine and advanced use of business intelligence (BI) systems by management accountants. *Information & Management*, 61(1), 103888. <https://doi.org/10.1016/j.im.2023.103888>
- Munir, A., Aved, A., & Blasch, E. (2022). Situational awareness: Techniques, challenges, and prospects. *AI*, 3(1), 55–77. <https://doi.org/10.3390/ai3010005>
- Mussina, K., Mamrayeva, D., & Lemanowicz, M. (2020). Digitalization in the tourism industry of Kazakhstan. *Bulletin of the Karaganda University Economy Series*, 99(3), 64–72. <https://doi.org/10.31489/2020ec3/64-72>
- Nelson, R. R., Todd, P. A., & Wixom, B. H. (2005). Antecedents of information and system quality: An empirical examination within the context of data warehousing. *Journal of Management Information Systems*, 21(4), 199–235. <https://doi.org/10.1080/07421222.2005.11045823>
- Nhan, V. K., Tam, L. T., Dung, H. T., & Vu, N. T. (2022). A conceptual model for studying the immersive mobile augmented reality application-enhanced experience. *Heliyon*, 8(8), 1–11. <https://doi.org/10.1016/j.heliyon.2022.e10141>
- Nookhao, S., & Kiattisin, S. (2023). Achieving a successful e-government: Determinants of behavioral intention from Thai citizens' perspective. *Heliyon*, 9(8), 1–21. <https://doi.org/10.1016/j.heliyon.2023.e18944>
- Norrie, M., & Signer, B. (2005). Overlaying paper maps with digital information services for tourists. In A. Frew (Ed.), *Information and communication technologies in tourism 2005* (pp. 23–33). Springer. https://doi.org/10.1007/3-211-27283-6_3
- Omran, W., Casais, B., & Ramos, R. F. (2025). Attributes of virtual and augmented reality tourism mobile applications predicting tourist behavioral engagement. *International Journal of Human-Computer Interaction*, 41(21), 13335–13348. <https://doi.org/10.1080/10447318.2025.2470293>
- Ou, S., Liu, T., & Liu, Y. (2025). Neural mechanisms of the impact of rotated terrain symbols on spatial representation in orienteers: Evidence from Eye-tracking and whole-brain fNIRS synchronization. *Behavioral Sciences*, 15(10), 1314. <https://doi.org/10.3390/bs15101314>
- Pabel, A., & Prideaux, B. (2016). Social media use in pre-trip planning by tourists visiting a small regional leisure destination. *Journal of Vacation Marketing*, 22(4), 335–348. <https://doi.org/10.1177/1356766715618998>
- Pál, M., & Albert, G. (2019). Digital cartography for geoheritage: Turning an analogue geotourist map into digital. In H. Fujita (Ed.), *Proceedings of the International Cartographic Association (ICA)* (Vol. 2, pp. 1–7). Copernicus Publications. <https://doi.org/10.5194/ica-proc-2-96-2019>
- Panda, D., Raje, G., & Ket, S. (2020). *Augmented reality based navigation and recognition system* [Paper presentation]. ITM Web of Conferences (Vol. 32, p. 03026). EDP Sciences. <https://doi.org/10.1051/itmconf/20203203026>
- Pang, J., Ye, J., & Zhang, X. (2023). Factors influencing users' willingness to use new energy vehicles. *PLOS One*, 18(5), e0285815. <https://doi.org/10.1371/journal.pone.0285815>
- Panou, C., Ragia, L., Dimelli, D., & Mania, K. (2018). An architecture for mobile outdoors augmented reality for cultural heritage. *ISPRS International Journal of Geo-Information*, 7(12), 463. <https://doi.org/10.3390/ijgi7120463>
- Papakostas, C., Troussas, C., Krouska, A., & Sgouropoulou, C. (2021). Measuring user experience, usability and interactivity of a personalized mobile augmented reality training system. *Sensors*, 21(11), 3888. <https://doi.org/10.3390/s21113888>
- Park, E., & Ohm, J. (2014). Factors influencing users' employment of mobile map services. *Telematics and Informatics*, 31(2), 253–265. <https://doi.org/10.1016/j.tele.2013.07.002>
- Paucher, R., & Turk, M. (2010). *Location-based augmented reality on mobile phones* [Paper presentation]. In 2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition – Workshops (pp. 9–16). IEEE. <https://doi.org/10.1109/CVPRW.2010.5543249>
- Penney, E. K., Agyei, J., Boadi, E. K., Abrokwhah, E., & Ofori-Boafo, R. (2021). Understanding factors that influence consumer intention to use mobile money services: An application of UTAUT2 with perceived risk and trust. *Sage Open*, 11(3), 21582440211023188. <https://doi.org/10.1177/21582440211023188>

- Perez Vallejos, E., Dowthwaite, L., Creswich, H., Portillo, V., Koene, A., Jirotko, M., McCarthy, A., & McAuley, D. (2021). The impact of algorithmic decision-making processes on young people's well-being. *Health Informatics Journal*, 27(1), 1460458220972750. <https://doi.org/10.1177/1460458220972750>
- Postma, A., Buda, D.-M., & Gugerell, K. (2017). The future of city tourism. *Journal of Tourism Futures*, 3(2), 95–101. <https://doi.org/10.1108/JTF-09-2017-067>
- Putrawangsa, D., Theresia, C., Yogasara, T., & Theopilus, Y. (2023). The effect of sex differences and experience of using virtual reality on presence. *Jurnal Optimasi Sistem Industri*, 22(1), 61–68. <https://doi.org/10.25077/josi.v22.n1.p61-68.2023>
- Putri, A., Rosmansyah, F. W., Triejunita, C. N., Sukmana, Y., Purnami, N., Sakina, S., Amalia, Y. S., & Rosmansyah, Y. (2025). An interactive mobile three-dimensional virtual laboratory for learning ear anatomy. *Smart Learning Environments*, 12(1), 19. <https://doi.org/10.1186/s40561-025-00371-8>
- Qiu, X., Yang, Z., Yang, J., Wang, Q., & Wang, D. (2024). Impact of AR navigation display methods on wayfinding performance and spatial knowledge acquisition. *International Journal of Human-Computer Interaction*, 40(10), 2676–2696. <https://doi.org/10.1080/10447318.2023.2169524>
- Rahimi, R., Hassan, A., & Tekin, O. (2017). Augmented reality apps for tourism destination promotion. In S. Rezaei (Ed.), *Apps management and E-commerce transactions in real-time* (pp. 236–251). IGI Global. <https://doi.org/10.4018/978-1-5225-2449-6.ch011>
- Rana, N. P., Dwivedi, Y. K., Williams, M. D., & Weerakkody, V. (2015). Investigating success of an e-government initiative: Validation of an integrated IS success model. *Information Systems Frontiers*, 17(1), 127–142. <https://doi.org/10.1007/s10796-014-9504-7>
- Ren, L., Yang, F., Gu, C., Sun, J., & Liu, Y. (2022). A study of factors influencing Chinese college students' intention of using metaverse technology for basketball learning: Extending the technology acceptance model. *Frontiers in Psychology*, 13(1), 1049972. <https://doi.org/10.3389/fpsyg.2022.1049972>
- Ren, M., Ma, Y., Fan, W., Li, M., & Feng, Y. (2022). An empirical study on the behavioral intention of college students' online ideological and political learning. *International Journal of Digital Multimedia Broadcasting*, 2022(1), 1–13. <https://doi.org/10.1155/2022/6349343>
- Rizaludin, M., Hadian, N., & Hayati, N. (2025). Integrating augmented reality with C4.5 algorithm to enhance tourism experience in Pekalongan. *Jurnal Ilmu Pengetahuan Dan Teknologi Komputer*, 10(4), 970–979. <https://doi.org/10.33480/jitk.v10i4.6244>
- Rohil, M. K., & Ashok, Y. (2022). Visualization of urban development 3D layout plans with augmented reality. *Results in Engineering*, 14(1), 100447. <https://doi.org/10.1016/j.rineng.2022.100447>
- Ruginski, I., Giudice, N., Creem-Regehr, S., & Ishikawa, T. (2022). Designing mobile spatial navigation systems from the user's perspective: An interdisciplinary review. *Spatial Cognition & Computation*, 22(1–2), 1–29. <https://doi.org/10.1080/13875868.2022.2053382>
- Ryu, M. H., Kim, J., & Kim, S. (2014). Factors affecting application developers' loyalty to mobile platforms. *Computers in Human Behavior*, 40(1), 78–85. <https://doi.org/10.1016/j.chb.2014.08.001>
- Sagnier, C., Loup-Escande, E., Lourdeaux, D., Thouvenin, I., & Valléry, G. (2020). User acceptance of virtual reality: An extended technology acceptance model. *International Journal of Human-Computer Interaction*, 36(11), 993–1007. <https://doi.org/10.1080/10447318.2019.1708612>
- Said, A., Ang, Z., & Iskandar, Y. H. P. (2023). Assessing consumers' acceptance of AR wayfinding for indoor shopping navigation in Singapore. *Wseas Transactions on Computer Research*, 12(1), 44–61. <https://doi.org/10.37394/232018.2024.12.4>
- Sasaki, R., & Yamamoto, K. (2021). Sightseeing navigation system from normal times to disaster outbreak times within urban tourist areas in Japan. *Applied Sciences*, 11(10), 4609. <https://doi.org/10.3390/app11104609>
- Schickler, M., Reichert, M., Geiger, P., Winkler, J., Funk, T., Weilbach, M., & Pryss, R. (2020). Flexible development of location-based mobile augmented reality applications with AREA: Implementation of a serious game shows the flexibility of AREA. *Journal of Ambient Intelligence and Humanized Computing*, 11(12), 5809–5824. <https://doi.org/10.1007/s12652-020-02094-9>
- Scholz, J., & Smith, A. N. (2016). Augmented reality: Designing immersive experiences that maximize consumer engagement. *Business Horizons*, 59(2), 149–161. <https://doi.org/10.1016/j.bushor.2015.10.003>
- Schroeder, T., Dodds, L., Georgiou, A., Gewald, H., & Siette, J. (2023). Older adults and new technology: Mapping review of the factors associated with older adults' intention to adopt digital technologies. *JMIR Aging*, 6(1), e44564. <https://doi.org/10.2196/44564>
- Shamsuddin, N. A. A., & Din, S. C. (2016). Spatial ability skills: A correlation between Augmented Reality (AR) and conventional way on wayfinding system. *Environment-Behaviour Proceedings Journal*, 1(2), 159–167. <https://doi.org/10.21834/e-bpj.v1i2.279>
- Shao, Z., Zhang, L., Chen, K., & Zhang, C. (2020). Examining user satisfaction and stickiness in social networking sites from a technology affordance lens: Uncovering the moderating effect of user experience. *Industrial Management & Data Systems*, 120(7), 1331–1360. <https://doi.org/10.1108/IMDS-11-2019-0614>
- Sobieraj, S., & Krämer, N. C. (2020). Similarities and differences between genders in the usage of computer with different levels of technological complexity. *Computers in Human Behavior*, 104(1), 106145. <https://doi.org/10.1016/j.chb.2019.09.021>

- Su, P.-Y., Hsiao, P.-W., & Fan, K.-K. (2023). Investigating the relationship between users' behavioral intentions and learning effects of VR system for sustainable tourism development. *Sustainability*, 15(9), 7277. <https://doi.org/10.3390/su15097277>
- Sun, H., & Zhang, P. (2006). Causal relationships between perceived enjoyment and perceived ease of use: An alternative approach. *Journal of the Association for Information Systems*, 7(1), 24. <https://doi.org/10.17705/1jais.00100>
- Sutherland, I. E. (1968). A head-mounted three-dimensional display. In American Federation of Information Processing Societies (Ed.), *Proceedings of the December 9–11, 1968, Fall Joint Computer Conference, Part I* (pp. 757–764). Association for Computing Machinery (ACM). <https://doi.org/10.1145/280811.281016>
- Tang, H., Shangguan, R., Zhao, J., Lin, J., & Fan, L. (2022). The influencing factors of college students' acceptance with VR implementation in aerobics teaching. *Journal of Physical Education*, 29(5), 132–138. <https://doi.org/10.16237/j.cnki.cn44-1404/g8.2022.05.012>
- Theodorou, C., Velisavljevic, V., Dyo, V., & Nonyelu, F. (2022). Visual SLAM algorithms and their application for AR, mapping, localization and wayfinding. *Array*, 15, 100222. <https://doi.org/10.1016/j.array.2022.100222>
- Tiwari, A., Pachori, R. B., & Sanjram, P. K. (2022). Isomorphic 2D/3D objects and saccadic characteristics in mental rotation. *Computers, Materials & Continua*, 70(1), 433–450. <https://doi.org/10.32604/cmc.2022.019256>
- Tom Dieck, M. C., Krey, N., & Cranmer, E. (2025). The value of augmented reality: Exploring hedonic and utilitarian augmented reality experiences. *Internet Research*, 35(3), 1252–1273. <https://doi.org/10.1108/INTR-11-2023-1048>
- Tran, T. T. M., Parker, C., Hoggenmüller, M., Hespanhol, L., & Tomitsch, M. (2023). Simulating wearable urban augmented reality experiences in VR: Lessons learnt from designing two future urban interfaces. *Multimodal Technologies and Interaction*, 7(2), 21. <https://doi.org/10.3390/mti7020021>
- Tyagi, P., Tyagi, P. K., Singh, A. K., Jain, E., & Singh, A. K. (2022). *Tourist experiences through mobile augmented reality* [Paper presentation]. 2022 International Conference on Electronics and Renewable Systems (ICEARS), (pp. 1605–1610). Institute of Electrical and Electronics Engineers (IEEE). <https://doi.org/10.1109/ICEARS53579.2022.9752325>
- Utari, U. A., & Fauziah, Y. (2023). Analysis of factors affecting intention to use and user satisfaction of paylater using DeLone & McLean Adoption Model. *Telematika*, 20(3), 343–461. <https://doi.org/10.31315/telematika.v20i3.10643>
- Veenendaal, B., Brovelli, M. A., & Li, S. (2017). Review of web mapping: Eras, trends and directions. *ISPRS International Journal of Geo-Information*, 6(10), 317. <https://doi.org/10.3390/ijgi6100317> [Cross0Ref]
- Venkatesh, V. (2000). Determinants of perceived ease of use: Integrating control, intrinsic motivation, and emotion into the technology acceptance model. *Information Systems Research*, 11(4), 342–365. <https://doi.org/10.1287/isre.11.4.342.11872>
- Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, 46(2), 186–204. <https://doi.org/10.1287/mnsc.46.2.186.11926>
- Wang, C., Xu, A., Sui, X., Hao, Y., Shi, Z., & Chen, Z. (2021). A seamless navigation system and applications for autonomous vehicles using a tightly coupled GNSS/UWB/INS/map integration scheme. *Remote Sensing*, 14(1), 27. <https://doi.org/10.3390/rs14010027>
- Wen, J., Deneka, A., Helton, W. S., Dünser, A., & Billinghamurst, M. (2014). *Fighting technology dumb down: Our cognitive capacity for effortful AR navigation tools* [Paper presentation]. Proceedings of the International Conference on Human-Computer Interaction (pp. 525–536). Springer. https://doi.org/10.1007/978-3-319-07227-2_50
- Werner, P. A. (2018). Review of implementation of augmented reality into the georeferenced analogue and digital maps and images. *Information*, 10(1), 12. <https://doi.org/10.3390/info10010012>
- Williams, D. R., Patterson, M. E., Roggenbuck, J. W., & Watson, A. E. (1992). Beyond the commodity metaphor: Examining emotional and symbolic attachment to place. *Leisure Sciences*, 14(1), 29–46. <https://doi.org/10.1080/01490409209513155>
- Windasari, N. A., Kusumawati, N., Larasati, N., & Amelia, R. P. (2022). Digital-only banking experience: Insights from gen Y and gen Z. *Journal of Innovation & Knowledge*, 7(2), 100170. <https://doi.org/10.1016/j.jik.2022.100170>
- Witmer, B. G., & Singer, M. J. (1998). Measuring presence in virtual environments: A presence questionnaire. *Presence: Teleoperators and Virtual Environments*, 7(3), 225–240. <https://doi.org/10.1162/105474698565686>
- Wyssenbach, T., Kaufmann, K., & Schwaninger, A. (2025). Segmentation influences learning: A study of knowledge acquisition through virtual reality and 2D video with airport security screeners. *Virtual Reality*, 29(3), 1–24. <https://doi.org/10.1007/s10055-025-01186-1>
- Xu, W., Ismail, M., Shahrudin, S., Quan, W., & Li, Y. (2025). Exploring tourists' intentions to adopt augmented reality in cultural heritage museums: Insights from a modified technology acceptance model. *Sage Open*, 15(2), 21582440251339936. <https://doi.org/10.1177/21582440251339936>
- Yahya, M. A., & Dahanayake, A. (2021). A needs-based augmented Reality system. *Applied Sciences*, 11(17), 7978. <https://doi.org/10.3390/app11177978>

- Yang, J.-M., Xiang, Y.-X., & Liu, C.-W. (2024). Enterprise sellers' satisfaction with business-to-business cross-border e-commerce platforms: Alibaba.com as an example. *Information Systems*, 122(1), 102348. <https://doi.org/10.1016/j.is.2024.102348>
- Yang, S. (2023). Storytelling and user experience in the cultural metaverse. *Heliyon*, 9(4), 1–14. <https://doi.org/10.1016/j.heliyon.2023.e14759>
- Yap, Y.-Y., Tan, S.-H., & Choon, S.-W. (2022). Elderly's intention to use technologies: A systematic literature review. *Heliyon*, 8(1), e08765. <https://doi.org/10.1016/j.heliyon.2022.e08765>
- Yen, H. Y., & Dong, Y. Y. (2025). Investigating user behavioral intentions and satisfaction in mobile games with cultural connotations through immersive experiences and the technology acceptance model. *Journal of the Science of Design*, 9(1), 1_21–1_30. https://doi.org/10.11247/jsd.9.1_1_21
- Yoo, J. (2020). The effects of perceived quality of augmented reality in mobile commerce—an application of the information systems success model. *Informatics*, 7(2), 14. <https://doi.org/10.3390/informatics7020014>
- Zakariya, K., Nizarudin, N. D., Jani, H. H. M., & Ibrahim, P. H. (2023). Potentials of interactive website in improving visitors' awareness on landscape heritage. *Asian People Journal*, 6(2), 136–154. <https://doi.org/10.37231/apj.2023.6.2.561>
- Zhang, B., Ali, K., & Kanesan, T. (2022). A model of extended technology acceptance for behavioral intention toward EVs with gender as a moderator. *Frontiers in Psychology*, 13(1), 1080414. <https://doi.org/10.3389/fpsyg.2022.1080414>
- Zhang, J., Chen, X., & Tan, M. (2020). Click or slide? Research on the effect of page-turning actions on users' online immersion intensity. *Foreign Economics & Management*, 42(2), 59–70. <https://doi.org/10.16538/j.cnki.fem.20191225.001>
- Zhang, J., Xia, X., Liu, R., & Li, N. (2021). Enhancing human indoor cognitive map development and wayfinding performance with immersive augmented reality-based navigation systems. *Advanced Engineering Informatics*, 50(1), 101432. <https://doi.org/10.1016/j.aei.2021.101432>
- Zhao, Y., Stefanucci, J., Creem-Regehr, S., & Bodenheimer, B. (2023). Evaluating augmented reality landmark cues and frame of reference displays with virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 29(5), 2710–2720. <https://doi.org/10.1109/tvcg.2023.3247078>
- Zhou, L., Xue, S., & Li, R. (2022). Extending the Technology Acceptance Model to explore students' intention to use an online education platform at a University in China. *Sage Open*, 12(1), 21582440221085259. <https://doi.org/10.1177/21582440221085259>
- Zhou, T., & Wang, C. (2016). Research on the continuous usage of knowledge community users based on social cognitive theory. *Journal of Modern Information*, 36(9), 82–87. <https://doi.org/10.3969/j.issn.1008-0821.2016.09.014>
- Zhou, X., Sun, Z., Xue, C., Lin, Y., & Zhang, J. (2019). *Mobile AR tourist attraction guide system design based on image recognition and user behavior* [Paper presentation]. Proceedings of the International Conference on Intelligent Human Systems Integration (pp. 858–863). Springer. https://doi.org/10.1007/978-3-030-11051-2_131
- Zhuang, X., Jiao, H., & Lu, K. (2022). Augmented reality interactive guide system and method for tourist attractions based on geographic location. *Journal of Electrical and Computer Engineering*, 2022(1), 1–13. <https://doi.org/10.1155/2022/7606289>
- Zou, N., Xu, Q., Wu, Y., Zhu, X., & Su, Y. (2023). An automated method for generating prefabs of AR map point symbols based on object detection model. *ISPRS International Journal of Geo-Information*, 12(11), 440. <https://doi.org/10.3390/ijgi12110440>
- Zou, Y., van Riel, A. C., Streukens, S., & Bloemer, J. (2023). Wanted! Investigating how elements from the personal usage context affect Gen Z consumers' value-in-use experience and engagement with mobile service. *Journal of Business Research*, 160(1), 113824. <https://doi.org/10.1016/j.jbusres.2023.113824>
- Утробина, Е. С., & Попов, В. П. (2021). Использование AR/VR технологий в мобильных картографических приложениях для развития сферы туризма [Use of AR/VR technologies in mobile cartographic applications for tourism development]. *Интерэкспо Гео-Сибирь*, 1(1), 211–217. <https://doi.org/10.33764/2618-981X-2021-1-211-217>

About the authors

Shuo Li is a student in the Department of Industrial Design at Central South University. Her research interests focus on interaction design, user experience, and product design.

Yuhui Wang is an associate professor of the industrial design department, Central South University. He is focused on usability theory, user experience and product design.

Lei Liu is a Professor in the Department of Industrial Design at Central South University. His research areas include product design and green design, with an emphasis on sustainable innovation and eco-friendly product development.